Report 3

Biomass-to-Jet: Using Gasification and Fischer–Tropsch Chemistry

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CHE 397: Senior Design II

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Table of Contents

1. Introduction

1.1 Problem Statement

Aviation is one of the most energy-intensive transportation modes and is inherently difficult to decarbonize due to strict safety regulations, long asset lifetimes, and the absence of cost-effective alternatives to high-density liquid fuels. Conventional jet fuel is derived from fossil crude oil through distillation and hydrotreating, and its combustion contributes significantly to global greenhouse gas emissions, as well as particulate matter and NOx, which impact both climate and air quality. Worldwide, aviation accounts for 2% of all carbon dioxide (CO_2) emissions and 12% of all CO_2 emissions from transportation. Global and national policy initiatives are establishing targets to decarbonize the aviation sector. ICAO's CORSIA program caps net CO_2 emissions from international aviation at 2020 levels through 2035, and the aviation industry has set a goal of reaching net-zero emissions by 2050. Because electrification and hydrogen are unlikely to be widely deployed in the near term, Sustainable Aviation Fuel (SAF) represents the most practical pathway to meet these objectives. In the U.S., the 2021 Sustainable Aviation Fuel Grand Challenge aims to scale domestic SAF production to 3 billion gallons annually by 2030 and 35 billion gallons annually by 2050, with a lifecycle reduction in GHG emissions of at least 50%, thereby creating long-term demand and policy support for technologies such as FT-SPK.

Sustainable Aviation Fuel (SAF) is an alternative fuel made from non-petroleum feedstocks such as biomass, waste oils, agricultural residues, and municipal solid waste. SAF has emerged as the most viable pathway for reducing aviation emissions because it is a drop-in fuel that can meet the same performance standards as Jet A without requiring modifications to jet engines, fuel systems, or airports. This is critical for near-term decarbonization because the physical and regulatory constraints of aviation make it impractical to implement rapid changes in propulsion systems, fuel storage, and aircraft design. SAF must therefore meet stringent performance specifications, including energy density, combustion characteristics, volatility, and cold-flow properties, while offering substantial lifecycle reductions in greenhouse gas emissions.

Multiple production pathways for SAF have been developed and approved under ASTM International standards, including Hydro processed Esters and Fatty Acids (HEFA), Alcohol-to-Jet (ATJ), and Fischer-Tropsch Synthetic Paraffinic Kerosene (FT-SPK). Among these, FT-SPK has gained significant attention due to its ability to convert a diverse range of low-value, sustainable feedstocks into high-quality paraffinic hydrocarbons with ultralow sulfur and aromatic content. When produced from biomass or waste-derived syngas, FT-SPK can provide substantial lifecycle emission reductions and additionally support waste diversion efforts, contributing both environmental and societal benefits.

Despite these advantages, SAF production remains limited by high capital intensity, feedstock logistics, and operational complexity, resulting in production volumes that comprise less than 1% of global jet fuel demand. However, the combination of international emissions mandates, national policy incentives, and long-term corporate sustainability commitments is driving rapid development of the SAF sector. In this context, evaluating the technical, economic, and environmental performance of FT-SPK production is essential to determine optimal process configurations, assess scalability, and identify barriers to commercialization. This project focuses on the design and preliminary analysis of a commercial-scale FT-SPK process capable of supporting the emerging demand for low-carbon aviation fuels.

1.2 Market Analysis

The Sustainable Aviation Fuel (SAF) market is poised for significant growth, driven by increasing environmental concerns, regulatory pressures, and advancements in fuel production technologies. Despite its promising potential, SAF production faces challenges, including high capital costs for establishing production facilities, difficulties in securing consistent feedstocks, and operational complexities. These factors contribute to the high production cost of SAF, which currently ranges from $3.50 to $6.00 per gallon, significantly higher than traditional jet fuel, which typically costs between $1.50 and $2.00 per gallon. This price disparity presents a barrier to widespread adoption. However, as production scales up, technology improves, and government incentives such as tax credits, subsidies, and other financial support are implemented, the price of SAF is expected to decrease. For example, the U.S. SAF Grand Challenge aims to produce 3 billion gallons of SAF annually by 2030, which will help lower production costs through economies of scale and infrastructure development.

The global SAF market is projected to grow from USD 2.7 billion in 2025 to USD 28.6 billion by 2032, at a CAGR of 39.95%. North America is expected to remain the largest market, accounting for 46% of the market share by 2024, largely due to strong government-backed initiatives such as the SAF Grand Challenge. Europe is also a key player, driven by the European Green Deal and national goals for carbon-neutral aviation. Meanwhile, the Asia-Pacific region is set to experience rapid growth, fueled by rising investments, increasing demand for cleaner fuels, and technological advancements in SAF production. The combination of these factors, alongside the scaling of SAF production, is expected to drive down costs and make SAF more competitive with traditional jet fuel, supporting its long-term adoption in the aviation industry.

1.3 Judging Criteria

The Fischer-Tropsch (FT) pathway was selected over alternative sustainable aviation fuel production pathways, based on its superior feedstock flexibility, blending capability, scalability, long-term feasibility for meeting the growing SAF demand, and its potential for reducing carbon emissions during the life cycle of the product.

One major advantage of FT is its ability to process a broad range of carbon-based feedstocks like woody biomass, agricultural, Municipal Solid Waste (MSW), etc. The design basis of the plant accounts for this advantage by choosing St. James Parish, Louisiana, where forestry residue is abundant at a low cost. In contrast, other competing processes like HEFA are constrained almost exclusively to rich lipid feedstocks which are highly limited. These feedstock sources are heavily competed for by the renewable diesel market, leaving little to no room for the SAF market. Alcohol to Jet (ATJ) feedstocks depend on sugar, starch, or cellulosic ethanol. Food and agricultural waste have a highly competitive market as these feeds are often used in biofuel industries to produce things like biodiesel. Because FT can utilize non-food, non-lipid, waste biomass at very large scales, it provides a more robust and sustainable feedstock supply chain.

Fisher-Tropsch derived synthetic paraffinic kerosene (FT-SPK), also known as jet fuel, is certified under the ASTM D7566 Annex A1 to be blended at up to 50 vol% with conventional Jet A/A-1 jet fuel. This high blending limit provides two critical advantages. It enables significant displacement of fossil fuel derived jet fuel per unit of SAF produced and allows for drop-in use without any modifications to aircraft engines, fuel handling systems, or airport logistics that are designed to operate with fossil fuel derived kerosene. Other pathways offer a lower maximum blend limit ranging from 10% to 30% thus FT offers both high blending capabilities and feedstock scalability.

The FT pathway also enables deep decarbonization because the biomass to liquids routes achieves significant carbon intensity reductions depending on feedstock logistics and hydrogen sourcing. Other pathways simply do not offer this level of decarbonization due to FT's ability to reduce greenhouse gas emissions.

The long-term expansion potential for FT-SPK is very promising as SAF initiatives and demands continue to increase. Existing commercial plants demonstrate that 10,000 barrels/day is sustainable and efficient making the design of this plant strategic. Competing processes are generally smaller in scale, are limited by supply of feedstocks, and are less aligned with long-term SAF volume requirements mandated by U.S and EU policy. The U.S. Gulf Coast also already supports biomass supply chains, gasification and syngas-handling experience, hydrogen production and contains existing refinery and upgrading systems.

In summary, the Fisher-Tropsch pathway was selected because it provides broad and abundant feedstock options, 50% blend certification, which matches or exceeds alternatives, superior scalability for large volume SAF production, and lower carbon intensity. For these reasons, FT to SAF offers the most robust, sustainable, and scalable pathway for meeting long-term aviation decarbonization targets compared to methods like HEFA or ATJ.

2. Process Description

2.1 Chosen Process

The general Fischer-Tropsch process to produce synthetic paraffinic kerosene, also known as Jet Fuel, is as follows. The first step is feedstock selection, which can be any carbon-based material known as biomass. In this case, wood-based biomass is gasified to create synthetic gas, syngas, which is primarily a mixture of carbon monoxide and hydrogen gas ($CO_{(g)}$ + $H_{2\,(g)}$). The syngas must then undergo cleaning to ensure that a relatively low concentration of carbon dioxide (CO_2) remains. The lean syngas then undergoes catalytic reactions in an FT reactor that uses cobalt as a catalyst in low temperatures to form long chain hydrocarbons. These hydrocarbons are then sent to upgrading and fractionation processes to produce synthetic paraffinic kerosene (SPK) that meets the ASTM D7566 standard. The final product once blended at up to 50% must meet the new ASTM D1655 standard.

The detailed process will follow the major sections of the block flow diagram highlighted below in (citation) which include biomass gasification and syngas cleanup, Fischer-Tropsch reactions, hydrotreating, and hydrocracking. The operating conditions are based on the process simulation modeled using ASPEN Plus v14

The process begins with the gasification of wood-based biomass to produce syngas which is then cleaned to produce lean syngas that are fed into the FT reactor. For this process, the necessary feed is found in the design basis section to be approximately 5,700 tonnes/day of biomass. Unlike fossil-based syngas production, biomass gasification produces syngas containing significant amounts of CO_2, particulates and ash, as well as other contaminants. Extensive gas cleanup and conditioning are required prior to FT synthesis.

The first step in the biomass gasification process, shown in Figure 1, is to heat and dry the incoming biomass in a dryer (*D-100*) operating at about 250°F to reduce the typical inlet moisture of 30-50 wt.% down to <10 wt.%. Air is fed into the dryer after being heated and it expels the dry biomass and excess moisture. The feed is then ready to be sent to the biomass gasifier (*R-100*) where the dried biomass along with oxygen is converted into raw syngas via partial oxidation reactions with oxygen. The primary reactions are shown below (Equation 1).

$$C+O_2 \rightarrow CO_2$$

$$C+\frac{1}{2}O_2 \rightarrow CO$$

$$C+H_2O \rightarrow CO+H_2$$

$$CO+H_2O \leftrightarrow CO_2+H_2$$

Equation 1: Primary Biomass Conversion Equations

Assuming a fluidized bed reactor is used then the temperature can be specified at ~1400°F with pressure an operating pressure of ~15 psia. The products of this reactor include raw syngas and tar.

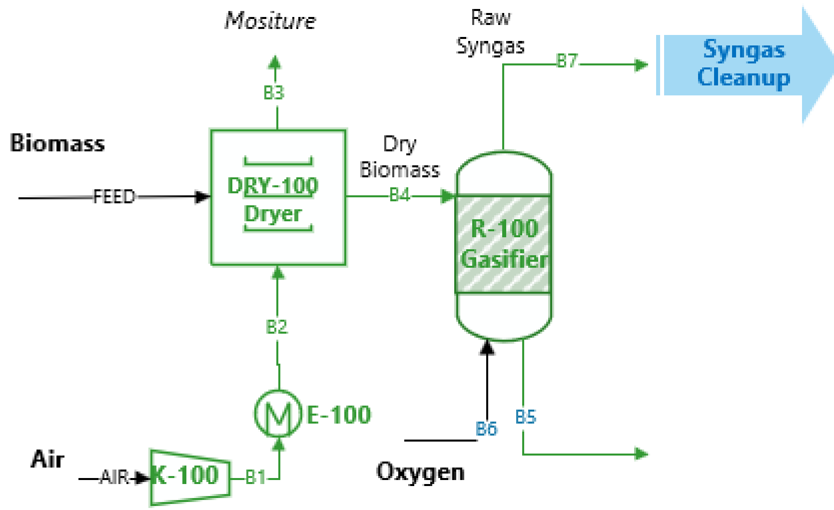

Figure 1: Biomass Gasification Process Flow Diagram

The cleanup process is as show in *Figure 2* below. Raw Syngas is fed into the cyclone *C-100* removes fine ash and char particles from raw syngas to protect the downstream equipment and operates at ~1400°F. The incoming gas will be compressed to a new pressure of 180 psia. E-100 then cools the hot syngas to enable downstream wet scrubbing and contaminant removal. Following cooling, the syngas scrubber *V-100* removes remaining contaminants using water and operates at 100°F and ~180pisa. The knockout drum *V-101* removes entrained liquid droplet and condensed liquids from scrubbed syngas at 86°F. The water-gas shift reactor *R-101* adjusts the H_2/CO ratio to approximately 1.7, as determined in the design basis and required for cobalt FT synthesis. The main reaction here is shown below (Equation 2). The reactor is expected to operate at high temperatures ~212°F) and at approximately atmospheric pressure 14.5 psia. From the design basis the steam to dry syngas ratio can be taken to be 0.5kg/kg.

$$CO + H_2O \leftrightarrow CO_2 + H_2$$

Equation 2: Water-Gas Shift Reaction

The final *V-102* absorber reduces the concentration of CO_2 to improve FT reactor productivity and operates at 145 psia removing 70-90% of CO_2. The resulting lean, conditioned syngas meets the FT feed specifications of having a H_2/CO ratio of 1.7, particulate concentration < 0.1 mg/Nm^3, and negligible concentration of sulfur and nitrogen species.

Key assumptions include that woody residues will have a consistent composition, air-blown gasification is acceptable, heat integration will greatly reduce external utility demands, and overall biomass to syngas efficiency aligns with reported ~70% cold gas efficiency.

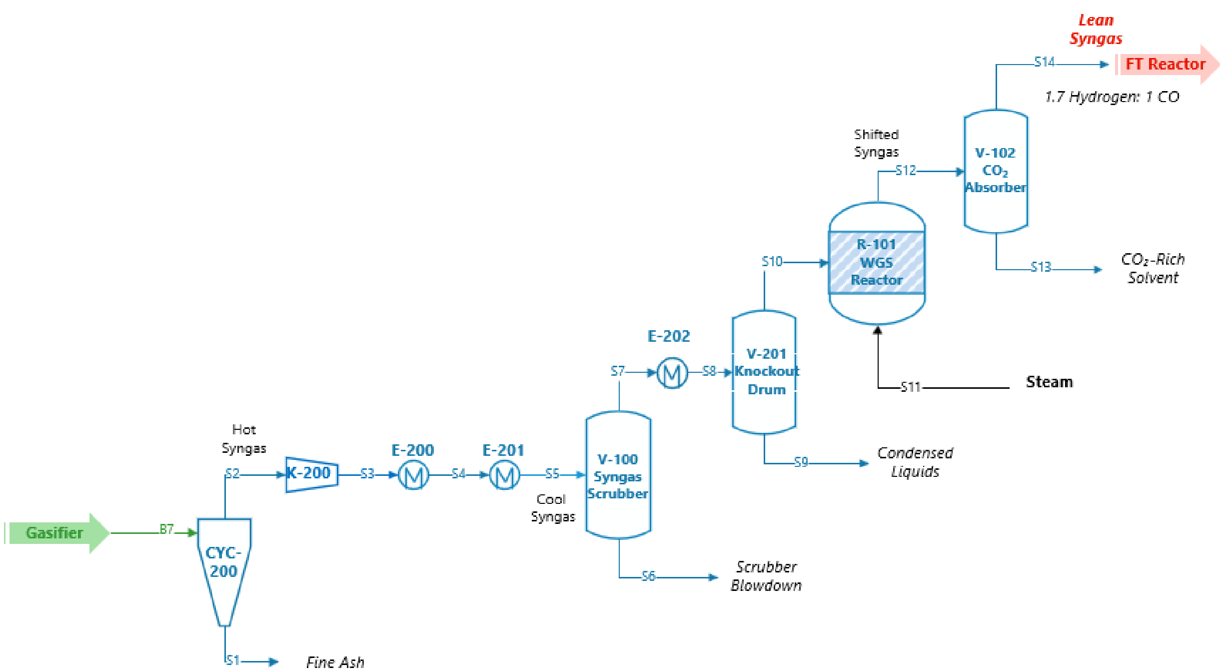

Figure 2: Syngas Cleanup Process Flow Diagram

Following the production of lean syngas, this is fed into the reactor section of the process *Figure 3*. This part of the process converts the conditioned syngas into long chain hydrocarbons producing waxes and middle distillates via the following chemical equations (Equation 3).

$$nCO+(2n+1)H_2 \rightarrow C_nH_{2n+2}+nH_2O$$

$$CO+H_2O \rightarrow H_2+CO_2$$

Equation 3: Fischer-Tropsch Reactions

The design basis will cover the optimal CO/H_2 ratio in the syngas feed. As previously mentioned, the FT-synthesis will occur in a fixed bed reactor at ~600°F and 290 psia with a 20 wt.% cobalt supported catalyst. The FT reactor produces hydrocarbons with a molecular weight distribution which can be described by the Anderson-Schulz-Flory distribution (Equation 4) .

$$\frac{W_n}{n}=1-\propto^2\propto^{n-1}$$

Equation 4:Anderson-Schulz-Flory distribution

The catalyst was chosen due to Cobalt’s high activity for long-chain paraffins in the desired SAF range of C_8-C_{16}. Alternatives not chosen include iron because it is less selective towards middle distillates, and nickel because of concerns of excessive methane formation. Key assumptions include those of the design basis.

High-pressure (HP) V-300 and low-pressure (LP) V-301 3-phase separators are added to separate heavy waxes (C20+) from liquid hydrocarbons and further separate liquid hydrocarbons into naphtha, middle distillates, and light hydrocarbons operating at ~300 psia and ~200 psia, respectively. Wax from the HP drum is sent to hydrocracking, and the fuel from the LP drum is vented. The LP drum *V-301* separates water and light hydrocarbons, removes inert gases, and vents flue gas. The light hydrocarbons are sent to the hydrocracking unit, and the upgrading processes follow.

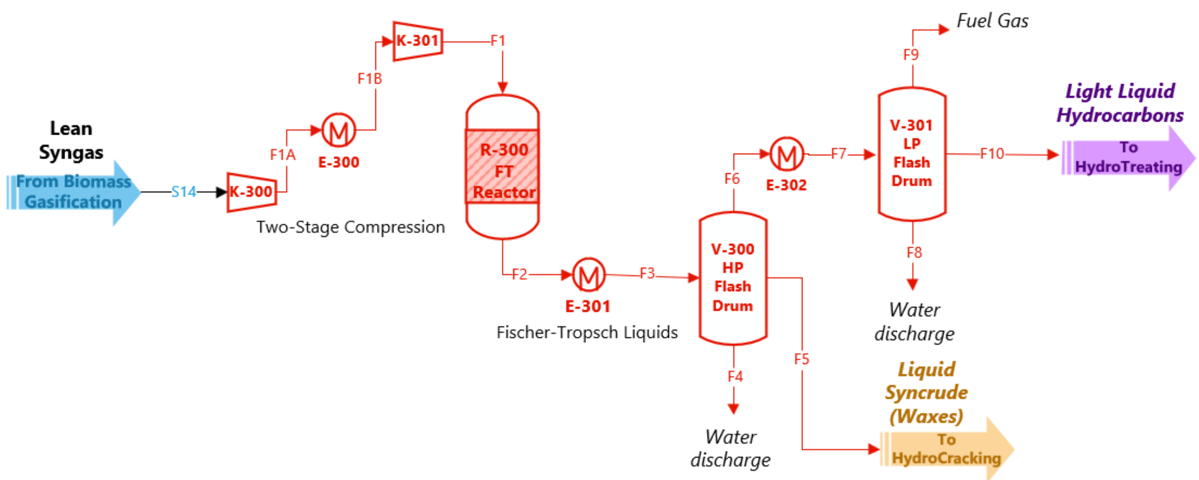

Figure 3: Fisher-Tropsch Synthesis Process Flow Diagram

The cobalt-catalyzed Fisher-Tropsch liquids which include linear paraffins, with a large fraction of heavy waxes (C_{20}^{+}) and a smaller fraction of middle distillates (C_9-C_{20}) must be sent to upgrading processes like hydrotreating and hydrocracking (Figure 4-Figure 5). While the hydrocarbons produced are sulfur and aromatics-free, they do not meet aviation fuel specifications due to excessive molecular weight and lack of isomerization. Hydrocracking and hydrotreating are required to crack FT waxes into jet fuel carbon range, isomerize linear paraffins into branched iso-paraffins to meet SAF freezing point limits, and produce co-product naphtha.

In the hydrotreating (*Figure 4*) section *R-400* serves as the primary hydrotreating reactor for FT liquids ensuring hydrogenation of olefins and removal of trace oxygenates. The feed includes FT light liquid hydrocarbons, and a blue hydrogen recycle stream. Unreacted hydrogen is recycled and supplied with purchased makeup hydrogen to comprise the hydrogen feed into the reactor. The feed is pumped and heated prior to entering *R-400*. Typical operating conditions include temperatures at ~450 °F and 900 psia. The flash drum *V-400* supplies unreacted hydrogen to the recycle stream and hydrotreated hydrocarbons into the first distillation column of the section. The primary fractionation column *C-400* separates light ends, naphtha, and unconverted waxes + jet fuel bottoms via atmospheric distillation. The column will typically operate at a top temperature of 150°C with a bottom temperature at 330-360°C at 1-2 bar. The final fractionation occurs in the finishing vacuum column *C-401* to separate SAF from heavier diesel-range hydrocarbons. Operating conditions here are like refinery kerosene fractionators and main products include SAF and diesel co-product.

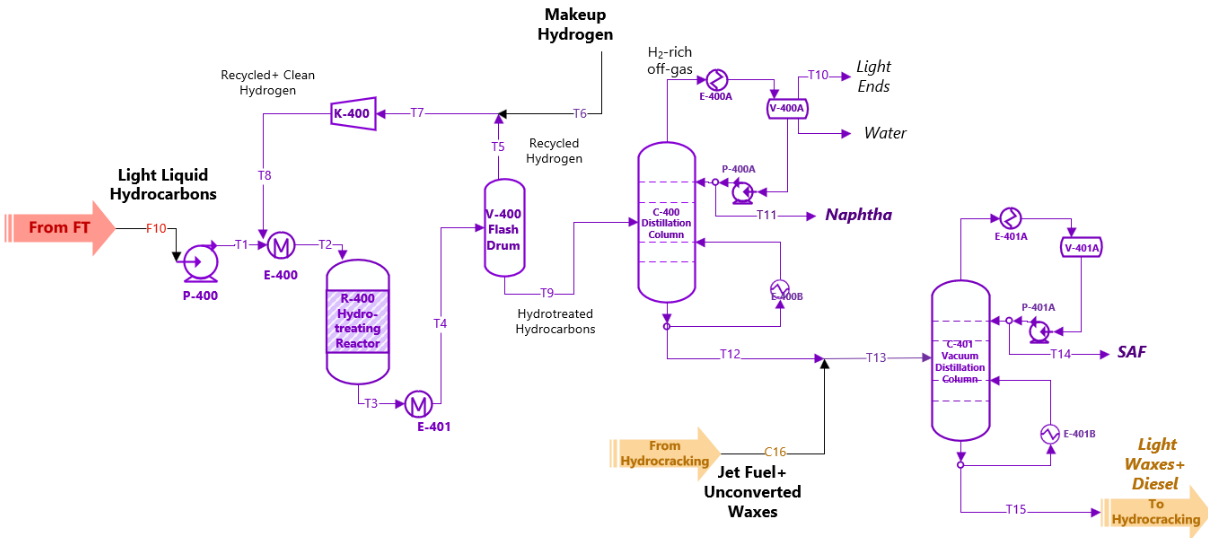

Figure 4: Hydrotreating Process Flow Diagram

The hydrocracking section (*Figure 5*) serves to crack FT wax into lighter paraffins while simultaneously isomerizing products. Feed streams include light waxes from hydrotreating (*T16*), liquid Syncrude from the FT synthesis, and makeup blue hydrogen (*C10*). A similar hydrogen gas recycle stream from the hydrotreating section is adopted here. The hydrocracker typically operates at 350°F and 900 psia. The conditions are chosen to promote isomerization and high wax conversion. The V-500 3-pase separating drum separates unreacted hydrogen, water discharge, and liquid cracked hydrocarbons at 120°F and 300 psia. The fractionation column *C-500* recovers Naphtha and light ends. Unconverted waxy material is recycled back to the hydrotreater to maximize SAF yield and overall carbon efficiency.

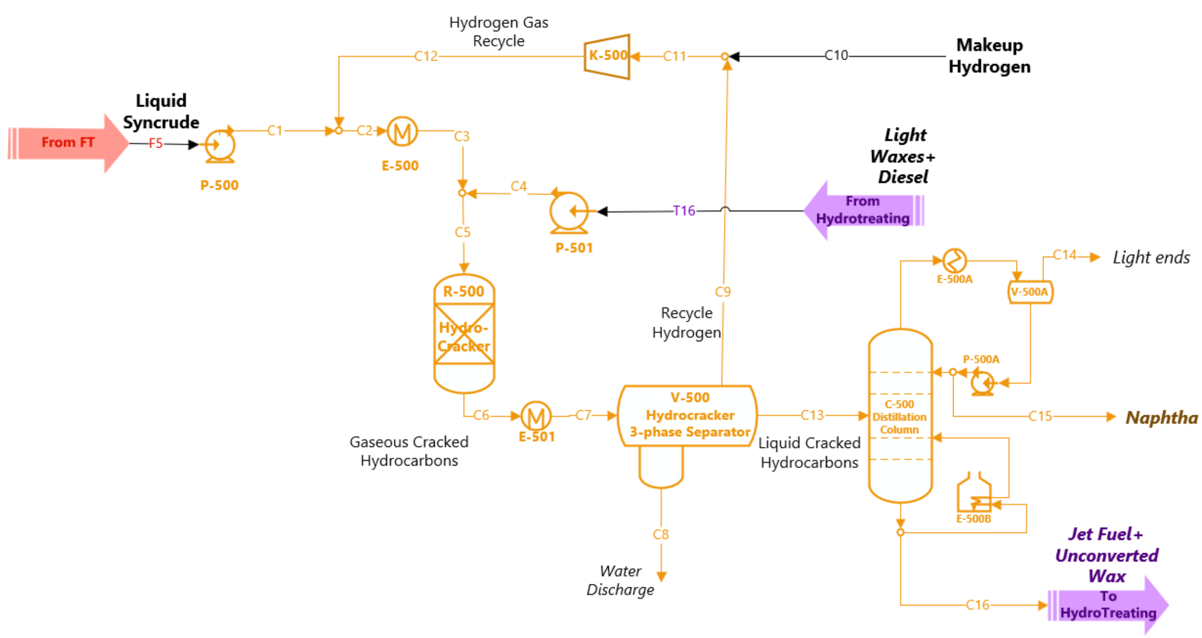

Figure 5: Hydrocracking Process Flow Diagram

Key assumptions in the upgrading section include that FT liquids contain negligible sulfur and nitrogen, eliminating the need for deep hydrotreating. The hydrogen is also assumed to be externally supplied from regional U.S Gulf Coat infrastructure. The hydrocracking severity is tuned to maximizing jet fuel range rather than gasoline.

2.2 Competing Processes

One of the main competing processes of Fischer-Tropsch process is the Hydro processed Esters and Fatty Acids (HEFA/HVO) process. Like the name suggests, hydro processed esters and fatty acids are the feed stock for this process with hydrotreated vegetable oil as another option. The first step of the process is pretreatment to remove any impurities. Pretreatment is crucial for SAFs especially since unlike traditional crude oil, it contains contaminants like phosphorus and metals. The pretreatment follows 5 steps: feedstock blending, straining, caustic treatment/water washing, impurity adsorption, and vacuum and filtration. Post pre-treatment hydrogenation and hydrodeoxygenation occur. Hydrogenation saturates any double bonds in the fatty acid chains, stabilizing them. Hydrodeoxygenation is where oxygen is removed via the creation of water (H_2O). Both processes are catalyzed, the first is typically by Pd/C, Pt/C, Ru,C or Raney NI for hydrogenation and Ni, Co, Mo, W and Al_2O_3 as typical catalysts for hydrodeoxygenation.

Following Hydrotreating, the product contains a large concentration of high carbon range materials C15- C18 paraffins, and this material is typically hydrocracked down to a useful range of carbons for SAFs (C8-C16). After the material is cracked there is a large concentration of straight chains that must be converted into branched isomers. The branched isomers have differing physical properties that are desirable for aviation fuel. These isomers control the fuel's ignition time, freezing point, viscosity, density, and flash point all of which must fall within aviation limits. After the isomers are created, they are separated via a high-pressure hot separator, cold high-pressure separator, and then fractionator columns. This leaves the products jet-fuel, diesel or naphtha. After this, any remaining gas product has hydrogen recovered and is then recycled back into the hydrogenation reactor.

The HEFA process is not without its limitations and shortcomings. Primarily it is more economical for these plants to develop renewable diesel over SAFs. Secondly, feedstocks have existing uses such as food production. And finally, it is more rigid in its production of carbon ranges, typically producing only C8-C16. FT synthesis is able to be to be tailored to a desired range of carbons for various uses. The block flow diagram for a HEFA process is shown below (Figure 6).

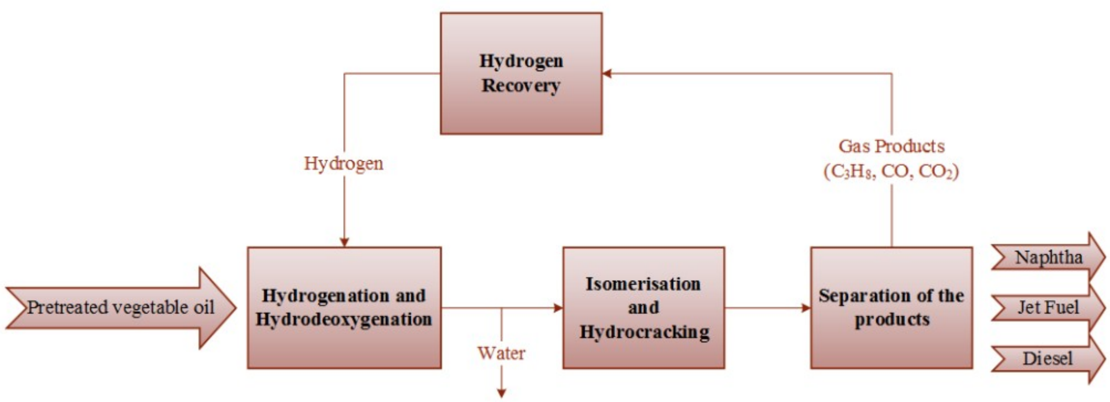

Figure 6: HEFA Block Flow Diagram

Another major competing process for Fischer-Tropsch synthesis is the alcohol to jet fuel (ATJ) route. The first step in the ethanol to jet process is the catalytic fueled dehydration of ethanol to ethylene. Typical catalysts for this process include HZSM-5 which is a zeolite and rare earth metals such as Lanthanum. HZSM-5 is the most common catalyst but is highly acidic, leaving it susceptible to coking. This results in a lower lifetime and large catalyst cost and regeneration rate. As a result, the ATJ process has a large catalyst cost and leaves room for improvements. Following the dehydration, the product is oligomerized. The olefins are then combined into larger carbon chains typically from the C8-16 range. Typical catalysts for this step include zeolites and Ni-based. The typical choice for catalysts is Ni-ALSBA-15 due to a large conversion of ethylene at 95% and a high selectivity for the C5-C7 range.

Following the oligomerization, the material is often sometimes separated before it is hydrogenated, but this step is not necessarily done in this way. The last step of the process is hydrogenation, and this process is typically done through two separate reactors with two different catalysts. The first reactor typically targets the shorter hydrocarbons in the C8-C11 range while the second reactor achieves a higher conversion of the larger hydrocarbon chains. Typical catalysts for the hydrogenation step include Ni-C and Pt/AL2O3. Following hydrogenation, the product is separated via distillation.

Similarly, HEFAs/HVOS alcohol to jet fuel has its limitations. It also is limited in its carbon ranges of C8-C16. It does offer a more flexible feedstock than HEFA but lacks a large commercial scale. It is limited by alcohol availability in each area as alcohol is a high demand product for other industries. Overall FT synthesis offers a large amount of flexibility in product creation and a large amount of feedstock options. For these reasons the decision to go with an FT-based production pathway was made. The block flow diagram for ATJ process is shown below (Figure 7).

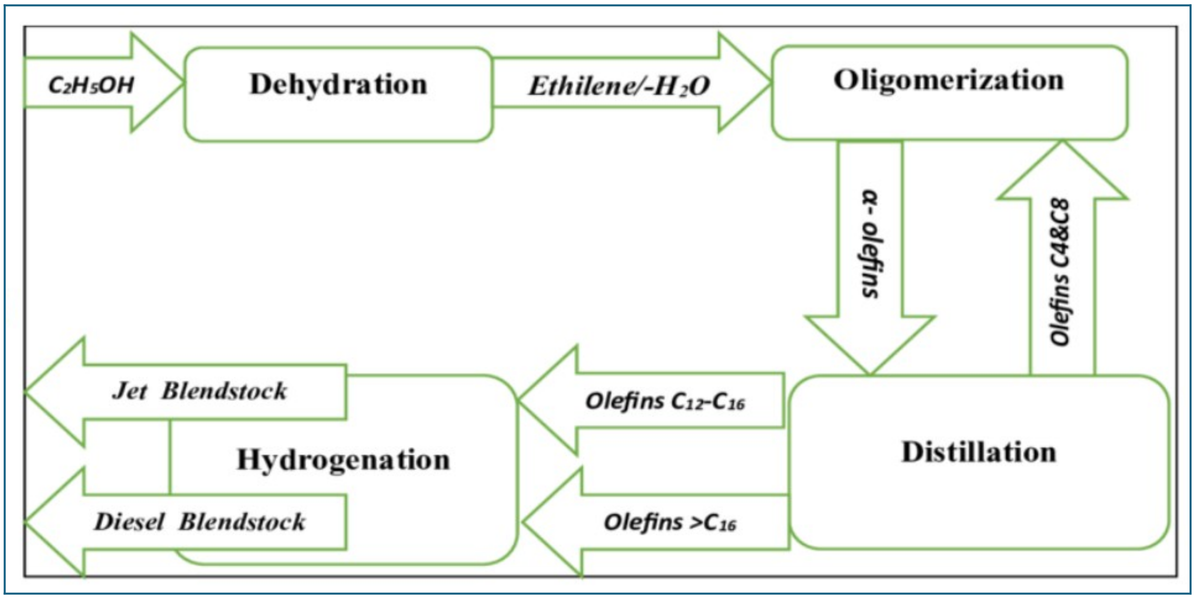

Figure 7: ATJ Block Flow Diagram

3. <u>**Process Control Analysis**</u>

3.1 Plant-Wide Control Strategy and Objective

To produce sustainable jet fuel safely and efficiently various controls must be put in place prior to plant operation. Each section highlighted in the BFD, Figure 14 has a different purpose therefore each section will be evaluated to apply the necessary controls. For this report only one unit, C-401, will be looked at in detail.

For the biomass gasification section of the process the main unit with operational concern is the gasifier. Variations with feed composition due to the inconsistent nature of biomass are the primary concern. Biomass will have varied particle sizes, moisture content and chemical composition that will impact syngas production. To combat disturbances in the feedstock, a typical process will control most of the following: feed flow, oxygen flow, ratio control, temperature and pressure. Since the output from the gasifier is temperature and pressure sensitive it is important to control these as well as possible. Deviations from the design conditions can lead to an increase of tar formation, ash content and increased methane formation. All these deviations lead to a lower quality of syngas and therefore lower quality SAF.

In the syngas cleanup section of the process the main topics of concern are syngas composition and tar removal. The most important unit operation in this section of the process is the cyclone. The cyclone will remove the ash from the system ensuring the rest of the undesired liquids can be removed in the subsequent flash drums producing the desired H2:CO ratio. Typical controls for the cyclone include inlet flow, pressure, temperature. These control variables directly correlate to the degree of separation in the cyclone and are therefore extremely important to maintain consistent operation. Another major area of concern for this area of the process is the water gas shift reactor. Since the water gas shift reaction is exothermic temperature control is important for safe operation. This alongside controlling the flowrate of steam into the reactor will for optimal syngas production.

In the FT section of the process the main area for concern is the Fischer Tropsch reactor itself. By far the most important control variable within the FT reactor is the temperature. Since the FT reaction is exothermic the temperature must be closely regulated to prevent a runaway reactor and ensure consistent products. The reactor pressure also has a strong correlation to the product distribution and conversion which must fall in a specific range for the system to operate efficiently. The system is designed to function primarily with a large number of heavier hydrocarbons and varying the temperature will change the carbon distribution greatly affecting operation.

In the hydrocracking section of the process the main areas of concern are the hydrocracker and the distillation column downstream of it. Since the hydrocracker is an overall exothermic process, temperature control is important for both operational safety and reaction kinetics.

Pressure control alongside make-up hydrogen flow rate are other major necessary controls. The distillation column will have controls similar to the fractionation column to be discussed in the control loop section.

In the hydrotreater section of the process the areas of concern are the hydrotreater reactor and it's associated temperature, pressure, flow and conversion alongside the operation of both distillation columns. The first distillation column C-400 will require temperature, pressure and level controls to ensure constant production of on-spec naphtha and continuous separation of SAF.

3.2 Control loops for Vacuum Column *(C-401)*

The fractionation vacuum column C-401 is chosen to design full control loop because it is where the main product from the system is obtained and therefore control of the unit is of the utmost importance. The aim of the control of this unit is to maintain the product, synthetic paraffinic kerosene, on spec while also maintaining safe operating conditions. This is accomplished through 5 control loops outlined below.

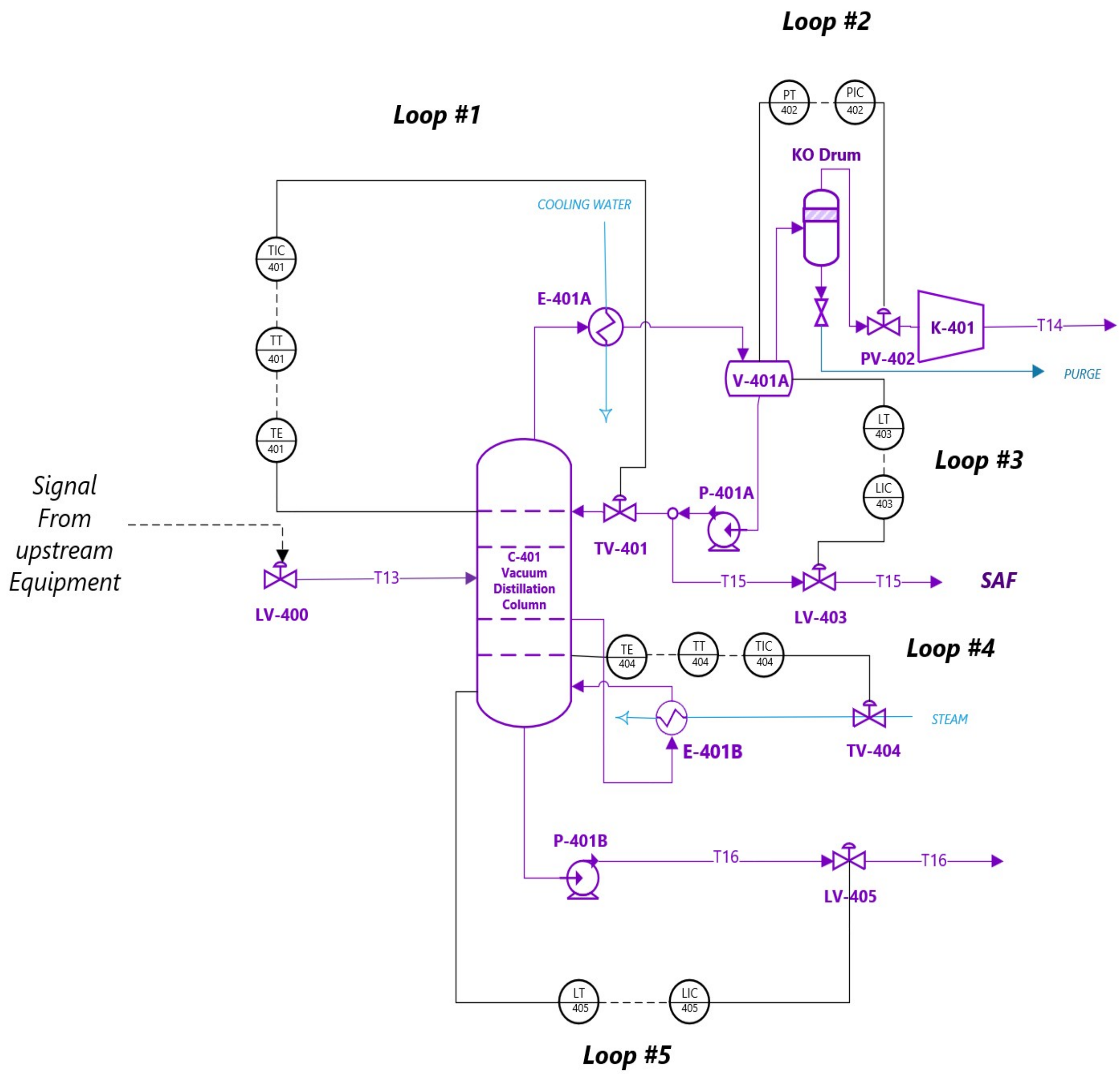

Figure 8*: Control Loop for C-401 before HAZOP Analysis*

The first control loop is designed to control the temperature near the top of the column. The temperature is recorded a few trays into the tower and if it deviates from the set-point the controller will send a signal to TV-401 to adjust the flowrate of the reflux entering the tower. If the temperature is lower than the set-point, the controller will reduce the reflux flowrate and if it is higher than the set-point, then it will increase the reflux flowrate.

The second control loop is designed to control the pressure in the column. The pressure is measured in the reflux-accumulator V-401A. If the pressure is higher than the set point the controller sends a signal to PV-402 to increase the flowrate of the gaseous stream allowing for a higher vacuum to be pulled. If the pressure is too low the controller will reduce the flowrate of the gaseous stream to reduce the vacuum. The low-pressure scenario is not as major of concern as the column is designed for full vacuum.

Next the third control loop is designed to maintain the level in the reflux-accumulator by manipulating the distillate flowrate. If the level is too high the controller sends a signal to LV-403 to increase the distillate flow rate. If the level is too low the controller sends a signal to reduce the distillate flowrate to return the level to its setpoint.

Additionally, the fourth control loop is designed to control the temperature in the reboiler. The temperature is once again measured a few trays from the bottom to help prevent rapid fluctuations. This temperature is measured and if it is higher than the set-point the controller sends a signal to TV-404 to reduce the steam flow rate. If the temperature is lower than the set-point, the controller sends a signal to increase the steam flow rate.

Finally, the fifth control loop is designed to maintain the level of the sump tank by manipulating the bottoms flow rate. If the sump level is too high the controller will send a signal to LV-405 to increase the bottoms flowrate. If the level is too low the controller will send a signal to decrease the bottoms flowrate to reach the set-point. These control loops are preliminary and will be modified post HAZOP analysis with additional safeguards.

4. Process Safety and Environmental Analysis

4.1 Chemical and Operational Safety

4.1.1 Material Safety

Material safety must be considered for each component present in the entire process. A general material safety description will be provided for each major section of the process as shown in the BFD in *Figure 14: Final Block Flow Diagram* along with a table highlighting individual components.

Biomass Gasification:

The biomass gasification section deals with the handling of woody biomass which can be susceptible to spontaneous combustion and must be handled and stored properly. The woodchips should be maintained in a low-dust cool environment. Respiratory protection against dust must be implemented when handling the biomass. Once the biomass is gasified, the resulting syngas must also be safely handled and its component's MSDS sheets should be reviewed. The components of syngas, carbon monoxide and hydrogen, are also reviewed.

Syngas Cleanup:

Once the syngas is ready to be sent to the cleanup section, a carbon dioxide absorber will ensure that a low concentration of CO_2 remains in the final syngas stream. The waste stream of the absorber will be a CO_2 rich solvent that must be disposed of adequately.

FT Synthesis:

Clean syngas will undergo various catalytic reactions to yield Fisher-Tropsch liquid hydrocarbons which are highly flammable and should maintain as far as possible from any possible ignition sources. This means installing ground containers to prevent static, explosion proof equipment, and following OSHA 1910.106 standards for flammable liquid storage and handling. The condensed liquids from the various cleanup units should be disposed of properly to prevent any contaminants coming into contact with the public.

Hydrotreating:

Following the low-pressure drum in the synthesis section of the process, light liquid hydrocarbons enter the hydrotreating section of the process. Similarly to the FT section, 1910.106 regulations are followed along with additional regulations for hydrogen like the 1910.103 standards.

Hydrocracking:

Comparably to the hydrotreating section, in the hydrocracking section the same regulations will be followed. OSHA also explicitly states that a process containing 10,000 lbs. of Hydrogen is PSM-covered as a flammable gas process. Hydrogen streams should be handled away from possible ignition sources.

4.1.2 Components

The individual components present in the entire process can be summarized in the table below. The following data was obtained from each component's MSDS sheet. The flammability limits are the highest and lowest concentrations in air that can propagate a flame if ignited at normal pressure and temperature. For substances that are not flammable, this property is not applicable like in the case of water. The boiling point is the temperature at which the substance can enter the liquid phase. While most substances have an agreed value, more complex hydrocarbons, like the products of this process, boil over a range of temperatures. The flash point is the lowest temperature of a liquid that can give off substantial flammable vapors. This chemical property can only be applied to substances that are liquid at standard conditions. The autoignition temperature is the lowest temperature at which a substance can spontaneously ignite without an external flame or spark. The toxicity of a substance is qualified as being potentially harmful to humans. The heat of combustion is the amount of heat produced when a substance undergoes complete combustion with oxygen. If the substance is not combustible, then the heat of reaction is given instead. All these parameters are important and must be accounted for when designing and planning for materials safety.

Hydrogen is an extremely flammable gas, classified as a level 4 fire hazard. The main health hazard is that hydrogen can quickly displace oxygen in air as it is a simple asphyxiant. It is non-toxic but dangerous due to its flammability and ability to lower oxygen levels untraceably.

Carbon monoxide is colorless and odorless highly toxic gases. It poses a level 3 health risk as it is toxic if inhaled and may cause damage to organs through prolonged exposure. It is also highly flammable, being a level 4 fire hazard, and should be kept in a cool dry area away from ignition sources. The OSHA Permissible Exposure Limit, PEL, is 50 ppm for workers.

Water is nonhazardous but when heated it can cause potential burns if it comes into contact with skin. Water and steam pose no health hazards but should be handled with care.

Carbon dioxide is an odorless, colorless asphyxiant.

Oxygen is a colorless, odorless gas at most temperatures and presents a combustion hazard especially at a large purity. Oxygen acts as an oxidizer and at the high concentrations necessary by the process it is highly combustible. It is important to keep flames and other ignition sources away from the feed oxygen streams.

Nitrogen is an inert gas that does not burn or support combustion. It poses a level 3 health risk because it is not poisonous but can be a simple asphyxiant. Liquid Nitrogen is also extremely cold and can cause instant frostbite upon contact with skin. Proper ventilation and cryogenic PPE when handling nitrogen is essential.

Hydrogen Sulfide is only present in trace amounts throughout the plant, but safety measures should be taken regardless. It is a colorless, flammable gas that poses a

Methane is also highly flammable and poses a suffocation risk. It must be handled in well ventilated areas away from ignition sources.

Ethane is highly flammable gas that poses explosion threats and must be kept in cool ventilated areas away from ignition sources

Propane is a colorless gas that is primarily a fire and asphyxiation hazard. It poses a health rating of 2 and a fire rating of 4, making it not as toxic but can ignite very easily. It should be kept in well-ventilated areas away from heat, sparks, flames, and other ignition sources. OSHA lists propane with a PEL of 1000 ppm as an 8-hour TWA and also notes it is treated as a simple asphyxiant, so oxygen deficiency is a major concern in confined or poorly ventilated spaces

Jet Fuel is a combustible petroleum liquid with a petroleum odor that is best treated as both a flammable liquid hazard and an inhalation / skin contact hazard. OSHA's kerosene (jet fuels) entry gives it an NFPA health rating of 2 and an NFPA fire rating of 2

Diesel fuel is a combustible petroleum liquid that presents more of a combustible liquid and chronic exposure hazard than an extreme flash-fire hazard under normal conditions. OSHA's diesel fuel page lists an NFPA health rating of 1 and an NFPA fire rating of 2.

Naphtha is a volatile petroleum liquid mixture that is both a flammability hazard and an inhalation hazard. OSHA's VM&P naphtha entry gives it an NFPA health rating of 1 and an NFPA fire rating of 3. It can ignite much easier than diesel or jet fuel

***Table 1**: Component Chemical Properties*

Component	Flammability Range	Boiling Point (°C)	Flash Point	Autoignition Temperature (°C)	Heat of Combustion or Reaction kJ/mol
H_2	4.1-74.2	-252.88	n/a	500	-286
CO	12.5-74.2	-191.5	n/a	609	-283
H_2O	n/a	100	n/a	n/a	n/a
CO_2	n/a	-78.5	n/a	n/a	-393.5
O_2	n/a	-182.96	n/a	n/a	-418
N_2	n/a	-195.8	n/a	n/a	941
H_2S	4.3-45	-60.3	n/a	260	519
CH_4	5.3-14	-161.5	n/a	540	802
C_2H_6	3-12.5	-88.6	n/a	515	1419
C_3H_8	2.3-9.5	-42.25	n/a	450	2029
Jet Fuel	0.7-5.6	150-250	37.8	210	~7000
Diesel	6.0-13.5	170-360	>38	>200	~8090
Naphtha	1.1-7.5	30-200	23	225	~3680

4.1.3 Operational Safety

For the general operation safety of, the plant 29 CFR 1910 standards will be followed

***Table 2**: Operational Safety Guidelines*

Unit	Safety Concerns	Prevention and Mitigation
Biomass Dryer	Fire or Explosion Conditions	Temperature monitoring and high temperature shutdown.
	Dust from woodchips	Install dust suppression, proper ventilation, and enforce proper respiration protection
Air Compressor	Overpressure	Discharge pressure protection
	Seal leaks	Routine seal inspections
Biomass Gasifier	High Temperatures	Temperature monitoring and high temperature shutdown.
	Flammable gasses	Gas leak detection
Water gas shift reactor	Temperature rise	Tight temperature monitoring

		across bed
	Overpressure	Pressure relief valves
	CO Leaks	Use CO detection around reactor area
CO_2 Absorber	CO_2-rich solvent leaks	Provide secondary containment for possible solvent leaks
	Column Flooding	Differential Pressure monitoring
Syngas Compressors	High-pressure syngas leaks	Pressure monitoring
	Flammable Gases	CO detectors
	Seal failure	Seal monitoring and leak detection
Fischer-Tropsch Reactor	Highly Exothermic Reaction	Temperature measurements across reactor
	CO and H_2 Leaks	Feed control, pressure relief to safe disposal system
	Runaway temperatures	High-high temperature emergency shutdown
High Pressure Flash Drum	High Pressures	Pressure relief and shutdown system, level control
Low Pressure Flash Drum	Flammable vapor released	Route off gas safely
Hydrotreating Reactor	High pressure hydrogen and hydrocarbon mixture	High-high temperature and pressure trips
	Exothermic reaction and hotspot formation	Temperature monitoring across bed
	Flammable gas leak and fire risk	Relief to safe disposal system
Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Flooding	DP monitoring
Vacuum Distillation Column	Vacuum Loss	Vacuum monitoring and alarms
Hydrocracker	High pressure hydrogen	High-high pressure alarm and trips
	Strong exothermic reaction-hotspot formation	Multiple temperature measurements
	Hydrogen leak and fire risk	Relief to safe disposal
3-phase separator	Hydrogen rich gas release	Closed vent handling
Recycle Hydrogen Compressor	Hydrogen leaks	Seal monitoring and leak detection
	Overheating	High temperature shutdowns
	Fire and explosion risk	Flammable gas detection

Hydrotreater Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Over head vapor release	Safe overhead vent handling

4.2 Process Safety and Environmental Analysis

4.2.1 Air Emissions

The main source of emissions for the process is CO_2

Material	*Amount (tonne/op-year)*	*Source*	
Carbon Dioxide	149,300		
Flue Gas	309,482	Low Pressure Flash Drum	
Light Ends		12,135	Hydrotreater distillation column
Light Ends	4,950	Hydrocracker Distillation Column	

4.2.2 Liquid Waste Treatment

Material	*Amount (tonne/op-year)*	*Source*
Water	712,500	Biomass Moisture
Tar	9,151	Biomass Gasifier
Scrubber Blowdown	19,800	Syngas Scrubber
Water Discharge	35,550	HP Flash Drum
Water Discharge	6,280	3-phase Separator

4.2.3 Solid Waste Treatment

Material	*Amount (tonne/op-year)*	*Source*

Ash	20,176	Syngas Cleanup Cyclone

4.3 Hazard and Operability Study *HAZOP* for Vacuum Column *C-401*

A comprehensive hazard and operability study was conducted for the vacuum column C-401 as shown in the table below. The software OPEN PHA was used to conduct the study. Deviations were evaluated for the severity of their consequences and appropriate safeguards and recommended actions were applied.

***Table 3**: Hazard and Operability Study*				
Main Vessel- ***Vacuum Distillation Column C-401*** Intention- to separate useful jet fuel hydrocarbons from light waxes and diesel hydrocarbons via vacuum distillation				
Deviation	Consequence	Cause	Safeguard	Recommended Action
MORE Pressure	Vessel mechanically fails due to overpressure and potential fatality.	TV-404 fails open, excess steam flow provides more heat to the reboiler than the overhead condenser system can remove. SAF components no longer condense, vacuum pump not rated for higher flow of vapor, pressure builds to MAWP, vessel failure, loss	- PT-402 readout in control room. - PSV-001 safety relief valve. - TT-401 readout in control room	- Add PAH-402 - Add TAH-401

		of process containment and potential fatality.		
LESS Pressure	Vessel fails due to high vacuum, collapse	PV-402 fails open, excess flow through K-401 vacuum pump, pressure drops below vacuum set-point, potential to fail column, potential fatality.	- Column is designed for full vacuum (FV).	- Add PAL-402
LESS Flow	No new consequences see high pressure	TV-401 fails closed, loss of reflux, excess heat to column, potential for overpressure to MAWP, vessel failure, loss of process containment and potential fatality.	- PT-402 readout in control room. - PSV-001 safety relief valve. - TT-401 readout in control room	- Add PAH-402 - Add TAH-401
MORE Flow	No new consequences, see high pressure	TV-404 fails open, causing excessive vapor traffic through the column	- TT-401 readout in the control - -PSV-001 safety relief	- Add PAH-402 - Add-TAH-401

			valve	
LESS Level	See high pressure for ultimate consequence. Evaluate seal fire here.	LV-403 fails open, excess material pumped from V-401A, loss of level, pump P-401A runs dry, equipment damage, potential seal fire, potential injury. Loss of reflux also occurs, leading to high pressure.	- TT-401 readout in control room	- Add TAH-401 - Install redundant level devices with alarms. - Install double mechanical seal with seal routed to flare, high pressure alarms on seal pot.
MORE Level	No new consequences, see low pressure	LV-403 fails closed, material accumulates in V-401A and be enrtrained towards K-401, potential damage to P-401A, loss of vacuum pressure	- PT-402 readout in the control room - KO drum	- Add LAH-403 - Consider adding LAHH to isolate K-401
LESS Temperature	No new consequences, see low pressure	TV-404 fails closed, TIC-404 reads low, or TT-404 reads high. TV-401	- PT-402 readout in the control room - TT-401	- Add TAL-401

		fails open		
MORE Temperature	No new consequences, see high pressure	TV-404 fails open, TIC-404 can read high, or TT-404 can read low, allowing excess steam to E-401B TV-401 fails closed, loss of reflux, excess heat to column, potential for overpressure to MAWP, vessel failure, loss of containment and potential fatality	- PT-402 readout in the control room - TT-401 readout in control room - PSV-400 pressure relief valve	- Add TAH-404 - Consider TAHH to isolate steam - Consider adding LAL to cooling water flow on E-401A

After conducting the HAZOP study, additional safety measures were added to the final P&ID shown below in *Figure 9*. Most notably, a safety pressure relief valve was added to the column along with alarms to indicate high and low deviations from various setpoints. These controls were added to the column to ensure safe operation.

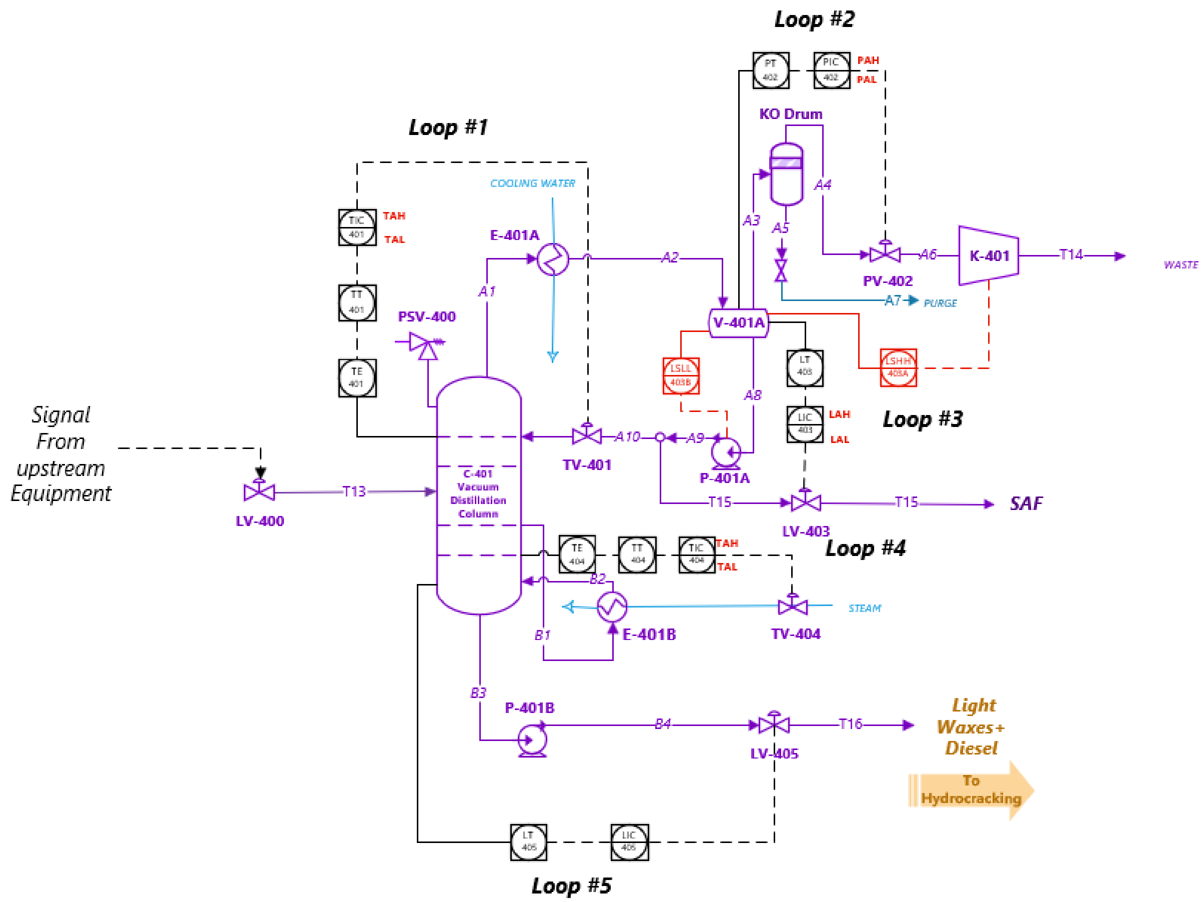

Figure 9: Final P&ID for Vacuum Fractionation Column C-401

5. Process Economic Analysis

An economic analysis was conducted to evaluate the financial feasibility of the proposed biomass-to-SAF process. This analysis includes estimates of both capital investment and operating costs associated with the facility. Capital costs were determined using a combination of Aspen Process Economic Analyzer results and manual equipment cost estimation methods, while operating costs were estimated based on raw material consumption, utilities, catalyst replacement, and fixed operating expenses. Using these cost estimates, a discounted cash flow analysis was performed to determine the discounted cash flow rate of return (DCFROR) for the project. The DCFROR was evaluated using both 5-year and 7-year MACRS depreciation schedules, and a sensitivity analysis was conducted to examine how variations in key economic parameters affect the overall profitability and feasibility of the process.

5.1 Capital Costs

The inside battery limits (ISBL) capital cost for the process was initially estimated using equipment cost results generated by Aspen. However, Aspen does not always provide accurate cost estimates for all equipment, particularly for custom equipment, reactors, or units outside its costing database. Therefore, major process units and any equipment that Aspen did not cost were manually estimated. These manual estimates were performed using the cost correlations provided in Table 7.2 of Towler and Sinnott, as well as by comparing the equipment size and duty to similar units reported in literature and scaling costs when Aspen estimates appeared unrealistic. The ISBL cost breakdown by section is shown in the figure below.

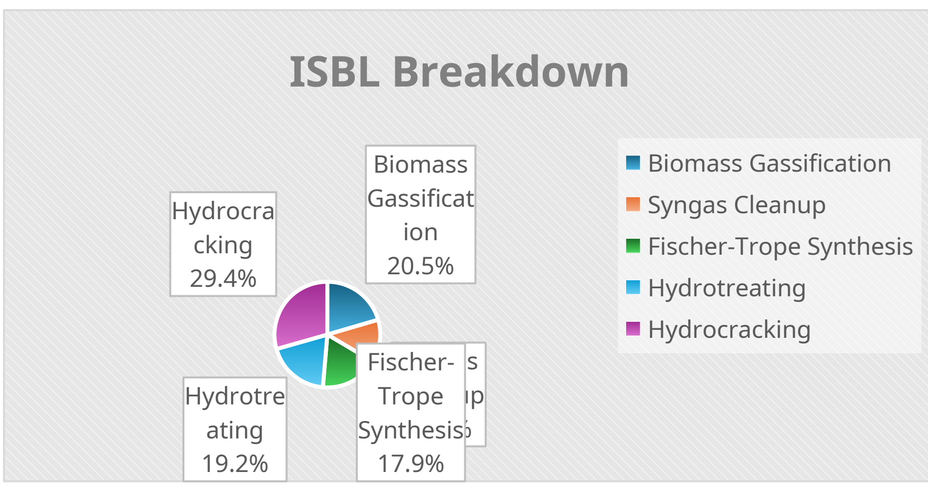

Figure 10: ISBL Breakdown by section

After estimating the base equipment costs, a material of construction correction factor F_m was applied to account for equipment constructed from alloys other than carbon steel. The material of construction for each unit is listed in Appendix V, and the corresponding F_m values were taken from Towler and Sinnott.

Next, a Chemical Engineering Plant Cost Index (CEPCI) correction was applied to adjust Aspen equipment costs to more recent cost conditions. Aspen's costing database corresponds to the year 2022 (CEPCI = 816). Costs were updated to 2025 conditions (CEPCI = 805) using the CEPCI ratio. However, the majority of major equipment costs were obtained from manual calculations based on more recent references, so this correction had only a minor impact on the overall capital estimate.

A location factor is typically applied to account for differences in construction costs between geographic regions. Since the plant is assumed to be located in Louisiana along the U.S. Gulf Coast, the standard Gulf Coast reference location was used and the location factor was taken as 1.0.

Once the ISBL cost was determined, additional capital costs were estimated using standard percentage factors. The following fractions of ISBL were applied:

- Outside Battery Limits (OSBL) = 0.40 × ISBL
- Engineering and Construction Costs (ECC) = 0.10 × ISBL
- Contingency (X) = 0.10 × ISBL
- Working Capital (WC) = 0.15 × ISBL

These factors account for offsite infrastructure, engineering and construction expenses, project uncertainty, and initial operating cash requirements. The resulting capital cost breakdown, including ISBL, OSBL, ECC, contingency, and working capital, is summarized in the table below. The sum of these components gives the total capital investment required for the plant.

Table 4: Capital Cost Summary	
Cost Type	Cost ($MM)
Inside Battery Limits (ISBL)	112.1
Outside Battery Limits (OSBL)	44.8
Engineering and Construction Costs (ECC)	11.2
Contingency (X)	11.2
Working Capital (WC)	16.8
Total Capital Cost (TCC)	**196.1**

5.2 Operating Costs

Operating costs were estimated by separating the variable cost of production (VCOP) and the fixed cost of production (FCOP). An 8000-hour operating year for the plant was assumed in all calculations. Variable costs include raw materials, utilities, catalyst replacement, and wastewater treatment. Fixed costs include labor, maintenance, and plant overhead. The cash cost of production (CCOP) is then determined as the sum of VCOP and FCOP:

$$CCOP = VCOP + FCOP$$

The calculated operating costs for the proposed process are summarized in the following subsections.

5.2.1 Variable Cost of Production (VCOP)

The variable cost of production consists of the costs of raw materials, utilities, catalysts, and wastewater treatment. These costs scale directly with plant throughput and therefore represent the largest portion of the operating expenses.

Utilities required for the process include cooling water, natural gas, and electricity. High-pressure and low-pressure steam are also used in the process. However, the plant generates significant quantities of steam through heat recovery in several heat exchangers.

The water–gas shift (WGS) reactor requires steam for operation. Rather than purchasing steam, the process uses steam generated internally through heat integration. Since the process produces more steam than is required, the excess steam offsets the cost of steam generation. Because the facility does not include infrastructure to export steam, the steam is assumed to be used internally at no net cost.

Utility prices used in the analysis were:

- Cooling water: $0.354/GJ
- Natural gas: $6/GJ
- Electricity: $0.0561/kWh

The total utility consumption and associated annual costs are summarized in the table below.

Table 5: Utility Consumption and Cost Summary

Utility	**Consumption**	**Unit**	**Annual Cost ($/yr)**
Cooling Water	150.5	GJ/	4.3×10^5

		hr	
Natural Gas	130	GJ/ hr	6.2×10^6
Electricity	1.83×10^4	kWh	8.2×10^6
Steam	Generated internally	—	—
Total Utility Cost	**—**	**—**	**$14.8 MM/yr**

Raw material costs represent the largest component of operating expenses. The primary feedstocks required for the process are biomass, oxygen, and green hydrogen.

Biomass provides the carbon source for gasification, oxygen is required for partial oxidation in the gasifier, and hydrogen is consumed in downstream upgrading reactions including hydrotreating and hydrocracking. The raw material consumption and annual costs are reported on the table below.

Table 6: Raw Material Consumption and Cost

Raw Material	**Flow Rate**	**Unit Price**	**Annual Cost ($/yr)**
Biomass	5.2×10^5 lb/hr	$0.03/lb	1.26×10^8
Hydrogen	1.6×10^3 kg/hr	$7.5/kg	9.55×10^7
Oxygen	5.0×10^3 kg/hr	$0.11/kg	4.4×10^6
Total Raw Material	**—**	**—**	**$225.7 MM/yr**

Cost			

Catalysts are required in the Fischer–Tropsch reactor, hydrotreating reactor, and hydrocracking reactor. The Fischer–Tropsch reactor uses a low-temperature iron catalyst promoted with Fe–Cu–K. Iron catalysts are commonly used for Fischer–Tropsch synthesis with biomass-derived syngas because they tolerate lower H_2/CO ratios and exhibit water–gas shift activity, which helps adjust the hydrogen balance during synthesis.

The hydrotreating and hydrocracking catalyst systems were selected based on the configuration described in the Rentech upgrading technical report, which outlines the catalyst loading schedule used for upgrading Fischer–Tropsch liquids. However, the detailed catalyst compositions are proprietary, so representative catalyst prices were used for economic estimation. The hydrotreater contains multiple catalyst layers consisting of a guard bed followed by hydrotreating catalysts. The guard bed, Crystaphase CatTrap 30, removes contaminants that could deactivate downstream catalysts. The remaining catalyst layers (HC-DM, HC-T, and AS-200LT) perform hydrogenation and stabilization reactions prior to hydrocracking.

The hydrocracker also contains multiple catalyst beds with different functions. The first section contains hydrogenation catalysts (HC-DM and HC-T), which saturate intermediate species and stabilize the feed. The second section contains HC-205LT, which provides the stronger cracking functionality required to convert heavier Fischer–Tropsch products into lighter liquid fuels. This split between hydrogenation and cracking catalyst beds is typical of industrial hydrocracking reactors and is consistent with the configuration described in the Rentech report.

The catalyst inventory for each reactor was estimated using the reactor volume and a typical catalyst bulk density. As an example, the catalyst requirement for the Fischer–Tropsch reactor was estimated using the relationship

$$m_{cat} = V_{reactor} \rho_{bulk}$$

Assuming a reactor catalyst volume of 18.4 m^3 and an iron catalyst bulk density of 800 kg/m^3, the catalyst inventory becomes

$$m_{cat} = (18.4\, m^4)(8000\ kg/m^3) \approx 14,750\ kg$$

Using an assumed catalyst price of $12/kg, the catalyst cost is 177,000$. Assuming a two-year catalyst lifetime, the annual catalyst replacement cost becomes

$$\frac{177,000}{2} = 88,500\ \$/yr$$

A similar procedure was used to estimate the catalyst inventory for the hydrotreating and hydrocracking reactors using their respective reactor volumes and catalyst packing densities. Because the hydrocracker contains separate hydrogenation and cracking catalyst sections, the total hydrocracker catalyst inventory was divided between these two catalyst types.

The resulting catalyst inventories and annual replacement costs are summarized in the table below.

Table 7: Catalyst Inventory and Annual Replacement Cost

Reactor	Catalyst System	Catalyst Mass (kg)	Cost ($/kg)	Lifetime (yr)	Annual Cost ($/yr)
FT Reactor	Fe–Cu–K promoted iron	14,750	12	2	88,500

	catalyst				
Hydrotreater	Guard + hydrotreating catalyst system (CatTrap 30, HC-DM, HC-T, AS-200LT)	7,060	25	4	44,125
Hydrocracker	Hydrogenation catalyst section (HC-DM / HC-T)	2672	45	4	30,000
Hydrocracker	Cracking catalyst section (HC-205LT)	6232	90	4	140,000
Total Annual Catalyst Cost	—	—	—	—	**300,000**

Wastewater generated from process condensate streams and cooling operations was estimated to be 2,500 gallons per hour. Wastewater treatment costs were estimated using an industrial treatment cost of $2 per 1000 gallons, resulting in an annual wastewater treatment cost of approximately $40,000 per year.

The total variable cost of production was calculated by summing the individual variable cost components:

$$VCOP = C_{raw} + C_{utilities} + C_{catalyst} + C_{wastewater} VCOP = 225.8 + 14.8 + 0.175 + 0.040$$

$$VCOP = 240.7\ \$\mathrm{MM/yr}$$

Raw materials account for the majority of the variable cost of production, representing approximately 88% of the total VCOP.

5.2.2 Fixed Cost of Production (FCOP)

The fixed cost of production (FCOP) represents operating expenses that do not vary significantly with plant throughput. These costs include operating labor, supervision, maintenance, plant overhead, and taxes and insurance. The FCOP was estimated using the standard cost factor approach described in Towler and Sinnott, where several cost components are estimated as fractions of labor cost or fixed capital investment.

The plant was assumed to require five operators per shift operating over three shifts, with an annual salary of $70,000 per operator. Additional costs for supervision, direct overhead, maintenance, plant overhead, and taxes and insurance were then estimated using typical percentage factors reported in Towler and Sinnott. The individual cost components used to calculate FCOP are summarized in Table X.

$$FCOP = C_{labor} + C_{supervision} + C_{direct\ overhead} + C_{maintenance} + C_{plant\ overhead} + C_{tax \wedge insurance}$$

The resulting total fixed cost of production was estimated to be $11.38 MM per year, as shown in the table below.

Table 8: Fixed Cost of Production Breakdown

Cost Component	**Basis**	**Annual Cost ($MM/yr)**
Operating Labor	15 operators × $70k/yr	1.05
Supervision	25% of labor	0.26
Direct Overhead	45% of labor + supervision	0.59
Maintenance	3% of ISBL	3.36
Plant Overhead	65% of labor +	2.69

	maintenance	
Taxes and Insurance	2% of fixed capital	8.66
Total FCOP	—	**11.38**

5.2.2 Cash Cost of Production (CCOP)

The cash cost of production (CCOP) represents the total annual operating cost required to operate the facility. CCOP is calculated as the sum of the variable cost of production (VCOP) and the fixed cost of production (FCOP).

$$CCOP = VCOP + FCOP$$

Using the previously calculated operating costs:

$$VCOP = 240.7\ \$\text{MM/yr},\ FCOP = 11.38\ \$\text{MM/yr}$$

the resulting cash cost of production becomes

$$CCOP = 240.7 + 11.38 = 252.1\ \$\text{MM/yr}$$

This value represents the total annual cost required to operate the plant and produce the final fuel products. The CCOP value will be used in the following section to evaluate plant profitability and determine the discounted cash flow rate of return (DCFROR). The operating cost breakdown is shown in the figure below.

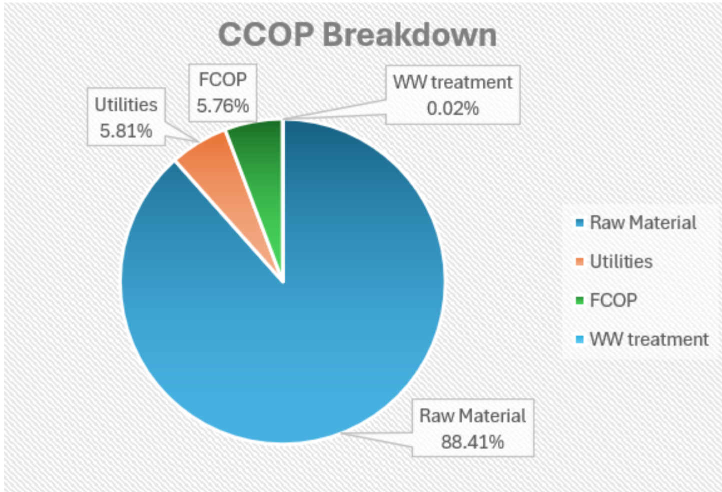

Figure 11: Operating Cost Breakdown (%)

5.3 Revenue and Gross Profit

Revenue for the proposed process was calculated from the sale of sustainable aviation fuel (SAF) and naphtha, together with the associated fuel credits. Market prices for the liquid fuel products were obtained from publicly available energy market data sources referenced in the bibliography. In addition to direct product sales, the process qualifies for fuel credits based on the carbon intensity of the produced fuels.

Because the process utilizes green hydrogen, the associated lifecycle CO_2 emissions are expected to be relatively low. As a result, the fuels produced by this process qualify for the higher end of available fuel credit values. Revenue was therefore calculated as the sum of product sales revenue and fuel credit revenue, as summarized in the following tables.

Table 9

: Annual Product Revenue

Product	**Production Rate**	**Price**	**Annual Revenue**

	(gal/hr)	($/gal)	($MM/yr)
SAF	17,680	3.1	438.5
Naphtha	6,028	2.8	135.0
Total Product Revenue	—	—	**573.5**

Table 10: Annual Fuel Credit Revenue		
Product	**Price ($/gal)**	**Annual Revenue ($MM/yr)**
SAF	1.4	198.0
Naphtha	0.8	31.8
Total Credit Revenue	—	**229.8**

The sum of product revenue and fuel credits gives the total annual revenue of the process, which amounted to 803.3 $MM/yr.

The gross profit of the process was determined by subtracting the cash cost of production (CCOP) from the total revenue. The results of this calculation are summarized in the table below.

Table 11: Profit Summary	
Category	**Value ($MM/yr)**
Total Revenue	803.3
Cash Cost of Production (CCOP)	252.1
Gross Profit	**551.2**

This gross profit value represents the annual earnings before depreciation, taxes, and financing considerations. These values are used in the following section to evaluate the economic feasibility of the process through a discounted cash flow rate of return (DCFROR) analysis.

5.4 Economic Feasibility and Discounted Cash Flow Analysis

The economic feasibility of the proposed process was evaluated using a discounted cash flow rate of return (DCFROR) analysis. The plant was assumed to have a 20-year operating lifetime, with an additional two years required for construction. During the construction period (years 1–2), the capital investment is distributed evenly and no revenue is generated. The plant is assumed to operate at 50% capacity during the first year of operation (year 3) before reaching full capacity for the remainder of the project lifetime.

As an initial measure of profitability, the simple payback period (PBP) and return on investment (ROI) were calculated using the total capital cost and annual gross profit determined in the previous sections. The simple payback period represents the time required for the project to recover the initial capital investment and is defined as

$$PBP = \frac{Total\,Capital\,Cost}{Gross\,Profit}$$

Using the calculated total capital cost and gross profit, the resulting payback period is approximately 0.36 years, corresponding to roughly four months of operation. The return on investment (ROI) is calculated as

$$ROI = \frac{Gross\,Profit}{Total\,Capital\,Cost} \times 100$$

which yields an ROI of approximately 281%, indicating strong profitability for the proposed process under the economic conditions considered.

A more rigorous evaluation was then performed using a discounted cash flow analysis, which accounts for the time value of money. The net present value (NPV) of the project is calculated as

$$NPV = \sum_{n=1}^{N} \frac{CF_n}{(1+i)^n}$$

where CF_n represents the annual cash flow in year n, i is the discount rate, and N is the project lifetime. A discount rate of 15% was selected to represent the opportunity cost of capital and the investment risk associated with large-scale chemical process facilities.

Annual pre-tax cash flow was calculated using

$$Pre\text{-}Tax \text{ Cash Flow } = \text{Revenue - VCOP - FCOP}$$

Taxes were applied to the taxable income, defined as the difference between the pre-tax cash flow and the depreciation of capital equipment:

$$Taxable \text{ Income} = Pre\text{-}Tax \text{ Cash Flow - Depreciation}$$

The tax payment was then determined using

$$Tax = (Taxable\,Income)(Tax\,Rate)$$

An effective tax rate of 26.5% was used for the analysis. This value represents the combined impact of the U.S. federal corporate tax rate (21%) and a typical Louisiana state corporate tax contribution, which is appropriate for a chemical facility operating in the Gulf Coast region.

Depreciation of capital equipment was calculated using the Modified Accelerated Cost Recovery System (MACRS). Both MACRS-5 and MACRS-7 depreciation schedules were evaluated. Depreciation was applied only to the capital investment excluding working capital, since working capital is recovered at the end of the project lifetime.

The DCFROR is defined as the interest rate at which the net present value of the project equals zero. This value was determined in Excel using the Goal Seek function by varying the interest rate until the final NPV became zero. The resulting DCFROR values are summarized below.

Table 12: DCFOR Summary

Depreciation Method	**DCFROR**
MACRS-5	**136.1%**
MACRS-7	**135.3%**

Both depreciation methods produce very high rates of return, indicating strong economic performance for the proposed plant. The MACRS-5 depreciation schedule results in a slightly higher return because a greater fraction of the capital investment is recovered earlier in the plant lifetime.

The unusually high DCFROR values can be attributed primarily to the high revenue generated from SAF fuel credits and favorable product prices, which significantly increase the annual profit of the process. Additionally, while capital costs were estimated using Aspen and manual calculations based on correlations from Towler and Sinnott, some equipment costs may still be underestimated due to limitations in the available cost correlations and the simplified geometric assumptions used for equipment sizing. As a result, the true capital cost of a commercial-scale facility may be higher than the value estimated in this analysis.

Detailed year-by-year calculations used to determine pre-tax cash flow, depreciation, taxable income, tax payments, discount factors, and NPV are provided in the Appendix.

5.5 Sensitivity Analysis

A sensitivity analysis was conducted to evaluate how variations in key economic parameters affect the overall profitability of the proposed process. The parameters analyzed include installed capital cost (ISBL), variable operating cost (VCOP), fixed operating cost (FCOP), and product revenue. Each parameter was varied between −30% and +30% of its base value, and the resulting net present value (NPV) was calculated using both the MACRS-5 and MACRS-7 depreciation methods. The resulting trends are shown in Figures 11 and 12, while the detailed numerical calculations are provided in the Appendix.

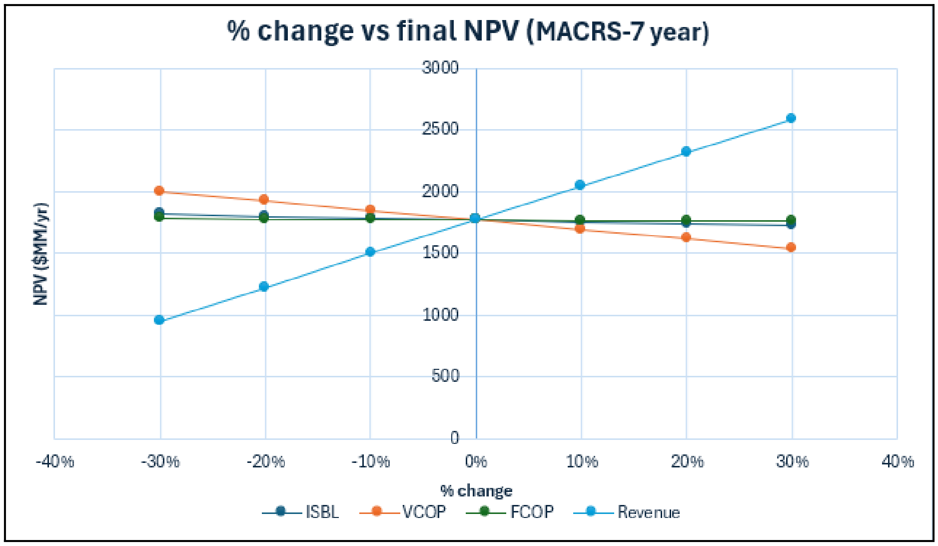

Figure 12: MACRS-7 year sensitivity analysis

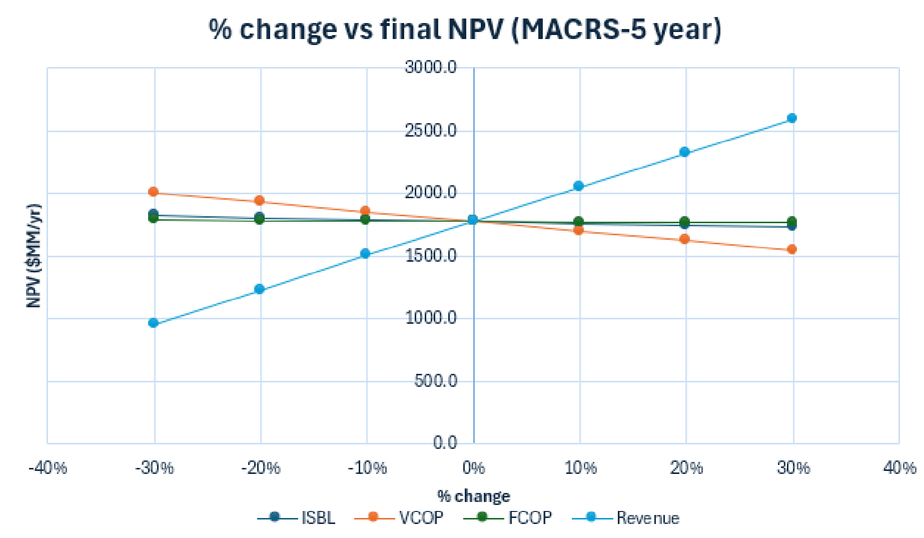

Figure 13: MACRS-5 year sensitivity analysis

The results show that product revenue has the strongest influence on project profitability. Increasing product prices significantly increases the NPV, while decreases in product price rapidly reduce profitability. This behavior occurs because revenue directly affects the annual cash flow of the plant, and therefore propagates through every year of the discounted cash flow calculation.

The variable operating cost (VCOP) is the second most influential parameter. Since VCOP represents the largest portion of annual operating expenses, changes in raw material or utility costs directly impact yearly profits and therefore alter the NPV. Increasing VCOP decreases the NPV, while decreasing operating costs improves profitability.

In contrast, the installed capital cost (ISBL) has a comparatively smaller impact on the final NPV. Although capital investment determines the initial negative cash flow during construction, this cost occurs only at the beginning of the project and does not affect annual operating profits. Because the plant generates substantial yearly revenue, the effect of the initial capital investment becomes less significant over the full project lifetime.

Similarly, fixed operating costs (FCOP) show only a minor effect on NPV. These costs represent a relatively small portion of the total annual expenses compared to raw materials and utilities, so variations in FCOP produce only modest changes in the overall profitability.

The sensitivity trends for MACRS-5 and MACRS-7 are nearly identical because depreciation primarily affects the timing of tax deductions rather than the overall cash generation of the plant. As a result, the same economic variables dominate profitability regardless of the depreciation schedule used.

The analysis indicates that the economic feasibility of the plant is most sensitive to product prices and variable operating costs, while capital cost and fixed operating expenses have a smaller influence on the final NPV. This behavior is expected for large-scale fuel production processes where long-term profitability is primarily driven by market prices and feedstock costs rather than the initial capital investment.

Overall, the economic analysis indicates that the proposed biomass-to-SAF process is highly profitable under the market conditions evaluated in this study. The combination of strong product revenues, policy-driven fuel credits, and relatively moderate operating costs results in large positive cash flows over the lifetime of the plant. Discounted cash flow analysis using both MACRS-5 and MACRS-7 depreciation schedules yields very high rates of return, with the MACRS-5 method providing a slightly more favorable economic outcome. Although some uncertainty remains in the capital cost estimates due to limitations of available cost correlations and simplified equipment geometries, the results consistently indicate that the proposed process is economically feasible and capable of generating substantial returns over its operating lifetime.

6. Appendix

Appendix I: Design Basis

This design is based on a ~10,000 barrels per day (bbl/day) sustainable aviation fuel (SAF) production facility using biomass gasification, syngas conditioning, low-temperature cobalt Fischer–Tropsch (FT) synthesis, and downstream hydrocracking/hydrotreating. The overall configuration and scale are benchmarked against commercial FT-SAF projects under development along the U.S. Gulf Coast—notably the DG Fuels plant in Louisiana, which will employ Johnson Matthey and bp’s FT CANS™ technology at a capacity of 13,000 bbl/day. This provides a realistic industrial reference point for sizing, performance expectations, and catalyst inventory at the scale considered in this study.

St. James Parish, Louisiana is selected as the plant location due to its favorable combination of biomass supply, industrial infrastructure, hydrogen availability, and proximity to emerging FT-SAF developments. Louisiana ranks among the top U.S. producers of forestry residues, logging waste, and wood-processing by-products, which form the primary biomass feedstock. The Mississippi River petrochemical corridor offers extensive utilities, existing pipelines, and refinery-adjacent infrastructure that reduce capital and operating costs for a biomass-to-liquids (BtL) facility. The region also benefits from strong hydrogen production capacity supported by ammonia and petrochemical operations, as well as growing SAF demand and supportive federal incentives such as the Inflation Reduction Act's SAF tax credits. Deep-water access through the ports of Baton Rouge and New Orleans simplifies logistics for catalyst delivery, equipment transport, and export of finished SAF.

To meet the target of 10,000 bbl/day SAF, the FT synthesis and upgrading section is designed for approximately 1,450 tonnes/day of total FT liquids, of which about 1,250 tonnes/day fall within the SAF carbon range (C_8–C_{16}), with the remainder as naphtha co-products. This production scale is consistent with biomass-to-liquids studies reporting overall biomass-to-liquids efficiencies of ~40% (LHV basis) for gasification followed by FT synthesis and upgrading.

At the front end of the process, the plant processes 5,700 tonnes/day of woody biomass, sourced from regional forestry residues and mill by-products. The biomass is dried on-site and fed to the gasifier to produce a raw syngas containing CO, H_2, CO_2, steam, and light hydrocarbons.

Biomass gasification naturally produces syngas with a low H_2/CO ratio, which is not suitable for low-temperature cobalt FT catalysts and FT synthesis in general. Because cobalt FT operation requires a target H_2/CO ratio near 2, the design incorporates a water–gas shift (WGS) reactor to partially convert CO and steam to H_2 and CO_2. Typical design studies report steam-to-syngas ratios between 0.4 and 0.7 kg steam per kg of dry syngas; here a value of 0.5 kg steam per kg of conditioned syngas is selected, resulting in a steam utility load of approximately 1,000 tonnes/day for syngas conditioning at the required FT reactor feed rate.

The FT synthesis section uses a low-temperature cobalt catalyst, selected for its high activity and strong selectivity toward long-chain linear paraffins, producing heavy wax (C_{20}^{+}) and middle distillate (C_9–C_{20}) hydrocarbons. Catalyst inventory is scaled from published industrial data indicating that a 100,000 bbl/day FT facility requires roughly 646 tonnes of cobalt. Normalizing to the capacity of this design yields an estimated 65 tonnes of cobalt, corresponding to approximately 320 tonnes of supported catalyst assuming a 20 wt% Co formulation.

Downstream of FT synthesis, the heavy wax and middle-distillate fractions are routed to hydrocracking and hydrotreating units to produce the desired blend of SAF and naphtha. The detailed hydrogen consumption associated with these upgrading steps is addressed separately in the mass balance section of the report.

Appendix II: Block Flow Diagram

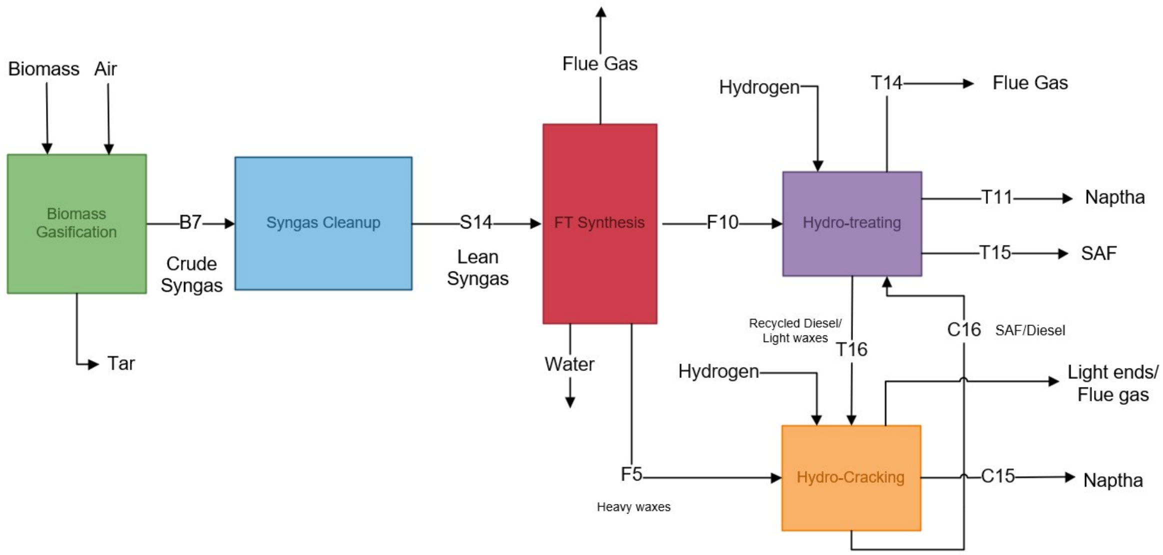

Figure 14: Final Block Flow Diagram

Appendix III: Process Flow Diagram

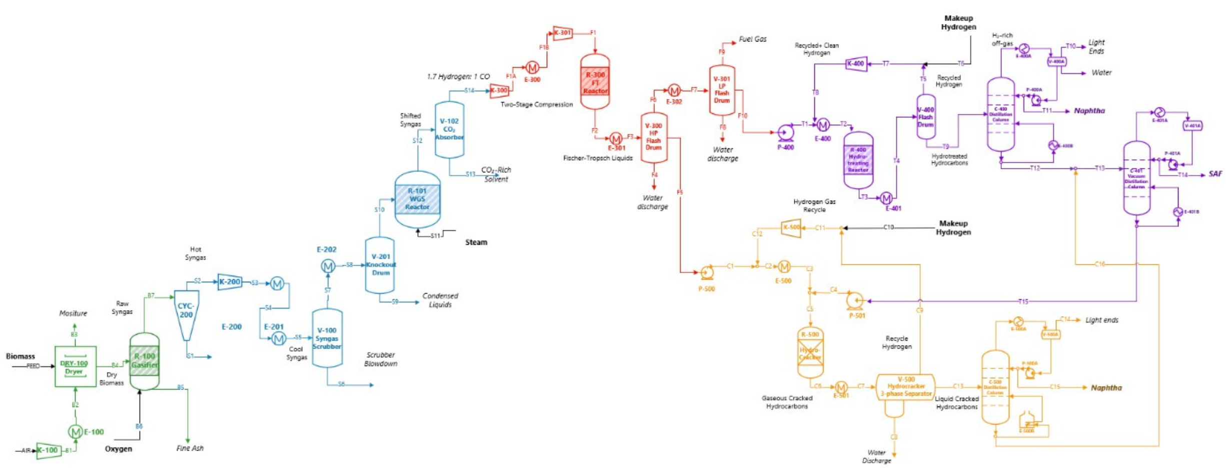

Figure 15: Final Process Flow Diagram

Appendix IV: Material Balance

The overall material balance is based on the final Aspen Plus v14 simulation and is organized by the major blocks in the flowsheet: Biomass Gasification (100), Syngas Cleanup/Conditioning (200), Fischer–Tropsch Synthesis (300), Hydrotreating (400), and Hydrocracking (500). The final configuration reflects the process changes made during optimization, most importantly the reduced biomass basis and the decision to eliminate diesel as an external co-product by routing diesel-range material and heavy wax back into the upgrading loop, so that they are ultimately converted into jet-range material.

The front-end biomass feed enters the process at approximately 523,598 lb/hr of dry woody biomass. After drying and the introduction of the oxidant, the gasifier produces a raw syngas stream B7 with a total flow rate of approximately 335,750 lb/hr. This stream contains approximately 261,959 lb/hr CO, 23,625 lb/hr H_2, 17,473 lb/hr CO_2, and 20,251 lb/hr CH_4, along with water vapor, minor light hydrocarbons, nitrogen, and approximately 5,044 lb/hr of ash. The cyclone removes the solid ash as stream S1, reducing the vapor-phase syngas flow to approximately 330,706 lb/hr in stream S2. No significant hydrocarbon loss occurs in this solids separation step, and carbon is conserved within the gas phase.

The syngas cleanup section conditions the gas for catalytic synthesis. After cooling, scrubbing, and condensation, water is partially removed in liquid knockout streams (S6 and S8), reducing the water loading prior to shift conversion. The water–gas shift reactor increases hydrogen concentration by converting CO, and the downstream CO2 removal unit separates CO2 into stream S13. Following these steps, the lean syngas stream S14 has a total mass flow of approximately 245,086 lb/hr. This stream contains approximately 218,321 lb/hr CO and 26,765 lb/hr H2, with CO2 reduced to negligible levels and no remaining solids. The reduction in total mass from the raw syngas basis reflects the removal of CO_2, condensed water, and solids. The S14 stream represents the total fresh syngas feed to the FT reactor before recycle blending.

In the Fischer–Tropsch section, the reactor feed F1 equals the conditioned syngas flow of approximately 245,086 lb/hr. Across the reactor, total mass is conserved; however, the species distribution shifts substantially. The reactor effluent F2/F3 includes approximately 33,111 lb/hr of reaction water formed, 34,118 lb/hr CO_2 generated or carried through, and multiple hydrocarbon fractions. Hydrocarbon products are represented in lumped carbon-number ranges: C5–C7 (~14,802 lb/hr), C8–C16 (~46,932 lb/hr), C17–C22 (~38,526 lb/hr), and C23–C40 (~23,975 lb/hr), with trace C42–C60 components. High- and low-pressure flash separation splits this effluent into a hydrocarbon liquid stream, F5 (approximately 110,182 lb/hr), and a gas stream, F6 (approximately 125,787 lb/hr). The gas stream contains unreacted CO and H_2, light hydrocarbons, CO_2, and water vapor and is partially recycled to maintain reactor utilization, with a controlled purge to prevent inert buildup. The liquid stream F5 contains the majority of the C8+ material and serves as the feed to the upgrading unit.

Hydrotreating receives the FT liquid stream F10, which represents the stabilized portion of F5 after initial separation, with a flow rate of approximately 18,754 lb/hr, as shown in the detailed stream table. Hydrogen addition and mild conversion redistribute minor quantities of light gases and water but do not significantly alter total hydrocarbon mass. Across the hydrotreating reactor and flash separation sequence (T1–T15), total hydrocarbon mass is conserved within expected hydrogen addition and purge limits. The stabilized liquid leaving the hydrotreating block is then fractionated to separate lighter naphtha-range components from heavier material directed to hydrocracking.

The hydrocracking section processes the heavy wax fraction originating from the FT separation (C23–C40 and C17–C22 material) along with heavier hydrotreated streams. The wax feed case represented by stream F6 entering the hydrocracking block shows approximately 26,222 lb/hr of heavy hydrocarbons prior to hydrogen addition. After mixing with hydrogen (C2/C3), the total reactor feed increases to approximately 51,095–52,430 lb/hr due to hydrogen incorporation. Within the hydrocracker, C20+ components are cracked and isomerized into lighter fractions, redistributing mass into C5–C7, C8–C16, and C17–C22 ranges while generating a light gas fraction. Reactor outlet separation removes a vapor-phase stream containing C1–C4 and light components, while the liquid phase proceeds to fractionation. Unconverted heavy material is internally recycled within the hydrocracking loop to increase overall conversion of wax into jet-range hydrocarbons.

When hydrotreated and hydrocracked liquid streams are combined, the total liquid production aligns with approximately 1,450 tonnes/day of finished fuels. Of this total, approximately 1,250 tonnes/day are recovered in the C8–C16 SAF range, with the balance primarily in the naphtha range. Diesel-range hydrocarbons are not withdrawn as a product stream in the final configuration but are internally recycled to the hydrocracking section. Overall mass closure across the entire flowsheet is maintained within Aspen's numerical tolerance, and the stream data confirm conservation of total mass through each major block, with expected redistribution among the vapor, liquid, and solid phases.

Table 13: Mass Balance Biomass Gasification Section-100										
Component	Units	Feed	Air	B1	B2	B3	B4	B5	B6	B7
Phase	Unitless	Solid	Vapor	Vapor	Vapor	Vapor	Solid	Solid	Vapor	Vapor
Temperature	0F	77	77	309	140	250	250	1472	77	1,397
Pressure	psia	14.7	19.7	48.7	43.7	14.7	14.7	15	15	15
Biomass	lb/hr	523,598	0	0	0	0	327,249	0	0	
CO	lb/hr	0	0	0	0	0	0	0	0	261,959
H_2	lb/hr	0	0	0	0	0	0	0	0	23,625
H_2O	lb/hr	0	0	0	0	196,349	0	0	0	7,003
CO_2	lb/hr	0	0	0	0	0	0	0	0	17,473
CH_4	lb/hr	0	0		0	0	0	0	0	20,251
C_2H_6	lb/hr	0	0	0	0	0	0	0	0	1
C_3H_8	lb/hr	0	0	0	0	0	0	0	0	trace
O_2	lb/hr	0	0	0	0	0	0	0	11,023	trace
N_2	lb/hr	0	0	0	0	0	0	0	0	393
Ash	lb/hr	0	0	0	0		0	0	0	5,044
Air	lb/hr		220	220	220	220	0	0	0	0
Tar	lb/hr	0	0	0	0	0	0	2,522	0	0
Total	**lb/hr**	**523,598**	**220**	**220**	**220**	**196,570**	**327,249**	**2,522**	**11,023**	**335,750**

Table 14: Mass Balance Syngas Section-200 Part 1								
Component	Units	B7	S1	S2	S3	S4	S5	S6
Phase	Unitless	Vapor	Solid	Vapor	Vapor	Vapor	Vap-Liq	Liquid
Temperature	^{0}F	1,397	1,397	1,397	3,166	500	100	100
Pressure	psia	15	15	15	185	180	175	180
CO	lb/hr	261,959	0	261,959	261,959	261,959	261,959	trace
H_2	lb/hr	23,625	0	23,625	23,625	23,625	23,625	trace
H_2O	lb/hr	7,003	0	7,003	7,003	7,003	7,003	4,958
CO_2	lb/hr	17,473	0	17,473	17,473	17,473	17,473	trace
CH_4	lb/hr	20,251	0	20,251	20,251	20,251	20,251	trace
C_2H_6	lb/hr	1	0	1	1	1	1	trace
C_3H_8	lb/hr	trace	0	trace	trace	trace	trace	trace
O_2	lb/hr	trace	trace	trace	trace	trace	trace	trace
N_2	lb/hr	393	0	393	393	393	393	0

Ash	lb/hr	5,044	5,044	0	0	0	0	0
Total	**lb/hr**	**335,750**	**5,044**	**330,706**	**330,706**	**330,706**	**330,706**	**4,958**

Table 15: Mass Balance Syngas Section-200 Part 2									
Component	Units	S7	S8	S9	S10	S11	S12	S13	S14
Phase	unitless	Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor	Vapor
Temperature	^{0}F	100	86	86	86	212	650	650	650
Pressure	psia	180	175	175	175	15	145	145	145
CO	lb/hr	261,959	261,959	trace	261,959	0	218,321	0	218,321
H_2	lb/hr	23,625	23,625	trace	23,625	0	26,765	0	26,765
H_2O	lb/hr	2,045	2,045	881	1,164	38,931	12,029	12,029	0
CO_2	lb/hr	17,473	17,473	trace	17,473	0	86,038	86,038	0
CH_4	lb/hr	20,251	20,251	trace	20,251	0	20,251	20,251	0
C_2H_6	lb/hr	1	1	trace	1	0	1	1	0
C_3H_8	lb/hr	trace	trace	trace	trace	0	trace	trace	0
O_2	lb/hr	trace	trace	trace	trace	trace	trace	trace	0
N_2	lb/hr	393	393	trace	393	0	393	393	0
Ash	lb/hr	0	0	0	0	0	0	0	0
Total	**lb/hr**	**325,747**	**325,747**	**881**	**324,866**	**38,931**	**363,797**	**118,711**	**245,086**

Table 16: Mass Balance FT Section-300 Part 1						
Component	Units	S13	F1	F2	F3	F4
Phase	Unitless	Vapor	Vapor	Vap-Liq	Vap-Liq	Liquid
Temperature	^{0}F	650	716	608	257	296
Pressure	psia	145	289	290	285	203
H_2	lb/hr	26,765	26,765	17	17	trace
CO	lb/hr	218,321	218,321	21,713	21,713	trace
H_2O	lb/hr	0	0	33,111	33,111	9,113
CO_2	lb/hr	0	0	34,118	34,118	4
H_2S	lb/hr	0	0	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	0	0	31,879	31,879	1
C_1-C_4	lb/hr	0	0	trace	trace	trace

C_5-C_7	lb/hr	0	0	14,802	14,802	trace
C_8-C_{16}	lb/hr	0	0	46,932	46,932	trace
C_{17}-C_{22}	lb/hr	0	0	38,526	38,526	trace
C_{23}-C_{40}	lb/hr	0	0	23,975	23,975	trace
C_{42}-C_{60}	lb/hr	0	0	13	13	trace
Total	**lb/hr**	**245,086**	**245,086**	**245,086**	**245,086**	**9,117**

Table 17: Mass Balance FT Section-300 Part 2

Component	Units	F5	F6	F7	F8	F9	F10
Phase	Unitless	Liquid	Vapor	Vap-Liq	Liquid	Vapor	Liquid
Temperature	^{0}F	296	296	80	122	122	122
Pressure	psia	203	203	198	300	300	300
H_2	lb/hr	trace	17	17	trace	17	trace
CO	lb/hr	115	21,598	21,598	trace	21,511	87
H_2O	lb/hr	2,030	21,967	21,967	21,567	279	122
CO_2	lb/hr	591	33,524	33,524	6	32,751	767
H_2S	lb/hr	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	1,354	30,525	30,525	trace	27,577	2,948
C_1-C_4	lb/hr	trace	trace	trace	trace	trace	trace
C_5-C_7	lb/hr	4,135	10,667	10,667	trace	3,078	7,589
C_8-C_{16}	lb/hr	39,566	7,367	7,367	trace	248	7,119
C_{17}-C_{22}	lb/hr	38,410	116	116	trace	trace	116
C_{23}-C_{40}	lb/hr	23,968	7	7	trace	trace	7
C_{42}-C_{60}	lb/hr	13	trace	trace	trace	trace	trace
Total	**lb/hr**	**110,182**	**125,787**	**125,787**	**21,573**	**85,460**	**18,754**

Table 18: Mass Balance Hydrotreating Section-400 Part 1										
Component	Units	F10	T1	T2	T3	T4	T5	T6	T7	T8
Phase	Unitless	Liquid	Liquid	Vapor	Vapor	Liquid	Vapor	Vapor	Vapor	Vapor
Temperature	^{0}F	122	131	572	608	122	122	77	101	445
Pressure	psia	300	865	860	865	860	354	100	100	900
H_2	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
CO	lb/hr	87	87	87	162	162	75	0	75	75
H_2O	lb/hr	122	122	122	123	123	1	0	1	1
CO_2	lb/hr	767	767	767	882	882	115	0	115	115
H_2S	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	2,948	2,948	2,948	3,045	3,045	97	0	97	97
C_1-C_4	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_5-C_7	lb/hr	7,589	7,589	7,589	7,600	7,600	11	0	11	11
C_8-C_{16}	lb/hr	7,119	7,119	7,119	7,120	7,120	1	0	1	1
C_{17}-C_{22}	lb/hr	116	116	116	116	116	trace	0	trace	trace
C_{23}-C_{44}	lb/hr	7	7	7	7	7	trace	0	trace	trace
Total	**lb/hr**	**18,754**	**18,754**	**18,754**	**19,054**	**19,054**	**300**	**0**	**300**	**300**

Total	**lb/hr**	**18,754**	**3,329**	**5,672**	**9,753**	**88,603**	**98,356**	**95,160**	**3,196**

Table 20: Mass Balance Hydrocracking Section-500 Part 1										
Component	Units	F6	C1	C2	C3	T15	C4	C5	C6	C7
Phase	Unitless	Liquid	Liquid	Vap-Liq	Vap-Liq	Liquid	Liquid	Vap-Liq	Liquid	Liquid
Temperature	^{0}F	3898	3898	3898	3898	0	0	3898	9179	9179
Pressure	psia	4456	4456	4456	4456	0	0	4456	9179	9179
H_2	lb/hr	trace	trace	5,276	5,276	trace	trace	5,276	9,179	9,179
CO	lb/hr	5,120	5,120	5,120	5,120	trace	trace	5,137	trace	trace
H_2O	lb/hr	4,648	4,648	4,648	4,648	trace	trace	4,649	9,179	9,179
CO_2	lb/hr	5,438	5,438	5,438	5,438	trace	trace	5,613	9,179	9,179
$H2_S$	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	26,289	26,289	26,289	26,289	trace	trace	27,863	trace	trace
C_1-C_4	lb/hr	trace	trace	18,124	18,124	trace	trace	19,398	1,254	1,254
C_5-C_7	lb/hr	8,206	8,206	8,206	8,206	trace	trace	8,208	0	0
C_8-C_{16}	lb/hr	3,828	3,828	3,828	3,828	0	0	3,828	0	0
C_{17}-C_{22}	lb/hr	246	246	246	246	0	0	246	0	0

C_{23}-C_{40}	lb/hr	38	38	38	38	0	0	38	0	0
C_{42}-C_{60}	lb/hr	-27,590	-27,414	-26,117	-24,782	trace	trace	-25,364	trace	trace
Total	**lb/hr**	**26,222**	**26,399**	**51,095**	**52,430**	**0**	**0**	**54,892**	**28,791**	**28,791**

Table 21: Mass Balance Hydrocracking Section-500 Part 2

Component	Units	C8	C9	C10	C10B	C11	C12	C13	C14	C15
Phase	Unitless	Liquid	Vapor	Vapor	Vapor	Vapor	Liquid	Vapor	Liquid	Liquid
Temperature	^{0}F	0	0	9179	0	0	9179	0	0	0
Pressure	psia	0	0	9179	0	0	9179	0	0	0
H_2	lb/hr	trace	0	9,179	0	0	9,179	0	0	trace
CO	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
H_2O	lb/hr	0	trace	trace	trace	trace	9,179	0	0	trace
CO_2	lb/hr	trace	0	trace	0	0	9,179	0	0	trace
$H2_S$	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4 (H^+)	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
C_1-C_4	lb/hr	trace	0	trace	0	0	1,254	0	0	trace
C_5-C_7	lb/hr	trace	0	trace	0	0	0	0	0	0
C_8-C_{16}	lb/hr	trace	trace	trace	trace	trace	0	trace	0	0
C_{17}-C_{22}	lb/hr	trace	trace	trace	trace	trace	0	trace	trace	0
C_{23}-C_{40}	lb/hr	trace	trace	trace	trace	trace	0	trace	trace	0
C_{42}-C_{60}	lb/hr	trace	trace	trace	trace	trace	trace	trace	trace	trace
Total	**lb/hr**	**0**	**0**	**9,179**	**0**	**0**	**28,791**	**0**	**0**	**0**

Appendix V: Equipment List

The following tables highlight the main equipment needed to run the process as well as the recommended material of construction.

Table 22: Biomass Gasification Section-100

Equipment	Description	Material of Construction
DRY-100	Biomass Drier	Carbon Steel
R-100	Gasifier	Carbon Steel

Table 23: Syngas Cleanup Section-200

Equipment	Description	Material of Construction
CYC-200	Syngas Cyclone	Carbon Steel

V-200	Syngas Scrubber	Carbon Steel
V-201	Knock Out Drum	Carbon Steel
R-201	Water Gas Shift Reactor	1.25Cr-0.5Mo Alloy
V-202	CO_2 Scrubber	Carbon Steel

Table 24: FT Reactor Section-300		
Equipment	Description	Material of construction
R-300	FT Reactor	316 S.S
V-300	HP Flash Drum	Carbon Steel
V-301	LP Flash Drum	Carbon Steel

Table 25: Hydrotreating Section-400		
Equipment	Description	Material of construction
R-400	Hydrotreater Reactor	1.25Cr-0.5Mo Alloy
V-400	Flash Drum	Carbon Steel
C-400	Distillation Column	Carbon Steel
C-401	Vacuum Distillation Column	Carbon Steel
P-400	Centrifugal Pump	Carbon Steel
P-400A	Centrifugal Pump	Carbon Steel
P-401A	Centrifugal Pump	Carbon Steel

Table 26: Hydrocracking Section		
Equipment	Description	Material of construction

R-500	Hydrocracker Reactor	2.255Cr-1.0 Mo Alloy
V-500	3-phase Separator	Carbon Steel
C-400	Distillation Column	Carbon Steel
P-500	Centrifugal Pump	Carbon Steel
P-501	Centrifugal Pump	Carbon Steel
P-500A	Centrifugal Pump	Carbon Steel

Appendix VI: Equipment Sizing

The following table displays the equipment specifications in order to meet the requirements of our plant's production.

Table 27: Heat Exchanger Duties

Equipment	Duty (KW)
E-100	5.33
E-101	13373
E-102	902
E-201	16976
E-300	9881
E-400	1864
E-401	2071
E-500	1341
E-501	4754

Table 28: Pump Flow Requirements

Unit	GPM
P-500	449.64

P-501	13.55
P-500 A	65.17
P-400	65.85
P-400A	12.23
P-401A	26.54

Table 29: Column Sizing

Distillation column	Diameter (ft)	Height (ft)	R.R	D/F
C-400	3	58	0.395	0.48
C-401	19.5	70	0.080	0.967
C-500	9	64	0.669	0.237

Appendix VII: Sample Hand Calculations

Hand calculations were performed to validate the Aspen RADFRAC design for the C-401 vacuum distillation column by estimating (1) number of stages, (2) column height, and (3) column diameter. Because the system is multicomponent (SAF/diesel-range hydrocarbons), the Fenske–Underwood–Gilliland (FUG) method was applied: the Fenske equation establishes a lower bound on the required stages at total reflux, the Underwood equations determine the minimum reflux condition consistent with the specified split, and the Gilliland correlation relates reflux to the practical number of stages required under operating conditions. The hand calculations were intended to provide a close comparison to Aspen trends rather than replicate a full rigorous stage-by-stage solution.

The vacuum distillation column (C-401) was modelled with a total condenser. Aspen data extracted for the analysis included feed/distillate/bottoms molar flow rates and mole fractions, feed K-values (for computing relative volatilities), the reflux flow $L$and distillate flow Dreported by RADFRAC, the feed thermal condition q, and tray-hydraulics quantities (internal vapor traffic and phase densities) from the Hydraulic Plots. Consistent with the Aspen split defined for SAF and diesel-range material, the light key (LK) was defined as C16 and the heavy key (HK) as C18. The feed quality was taken as $q = 0.96$, indicating a predominantly liquid feed entering the column.

The minimum number of theoretical stages was estimated using the Fenske equation, which provides the minimum stage requirement under the ideal limiting case of total reflux:

$$N_{\min} = \frac{\ln\left[\left(\frac{x_{LK}}{x_{HK}}\right)_{dist} \Big/ \left(\frac{x_{LK}}{x_{HK}}\right)_{bot}\right]}{\ln\left(a_{LK/HK}\right)} \quad \Rightarrow \quad N_{\min} = 4.0 \text{ theoretical stages}$$

A small N_{min} indicates that the specified LK/HK cut is feasible from a volatility standpoint and does not require an exceptionally tall tower under ideal conditions.

To apply the multicomponent Underwood method, relative volatilities were computed using Aspen feed K-values referenced to the heavy key:

$$a_i = \frac{K_i}{K_{HK}}$$

Feed K-values were used as a representative basis because the ordering and relative behavior of K across the component slate remains consistent, even though absolute K-values vary by stage temperature and pressure.

The Underwood root equation was then solved for ϕ using the feed mole fractions z_i, the computed α_i, and $q=0.96$:

$$\sum_i \frac{q\, z_i}{a_i - \phi} = 1 \qquad \Rightarrow \qquad \phi = 1.9982$$

The solution was confirmed to fall within the expected bracket for an HK-referenced volatility set, $1<\phi<\alpha_{LK}$. The minimum vapor rate was computed using the second Underwood equation:

$$V_{\min} = \sum_i \frac{a_i\, D\, x_{D,i}}{a_i - \phi} \qquad \Rightarrow \qquad V_{\min} = 661.7 \text{ lbmol/hr}$$

Considering that a total condenser is present, the minimum reflux flow was obtained from:

$$L_{\min} = V_{\min} - D \qquad \Rightarrow \qquad L_{\min} = 47.6 \text{ lbmol/hr}$$

with the distillate flow taken from Aspen as $D = 613.57 \text{ lbmol/hr}$, the corresponding minimum reflux ratio was:

$$\left(\frac{L}{D}\right)_{\min} = \frac{L_{\min}}{D} \qquad \Rightarrow \qquad \left(\frac{L}{D}\right)_{\min} = 0.0776$$

The operating reflux ratio was taken from Aspen RADFRAC results:

$$\frac{L}{D} = \frac{49.35}{613.57} \quad \Rightarrow \quad \frac{L}{D} = 0.0804$$

The operating reflux ratio being only slightly above $\left(L/D\right)_{min}$ indicates operation near minimum reflux, which is consistent with requiring a larger number of stages than the minimum-stage limit.

The Gilliland correlation was then used to estimate the actual number of theoretical stages at the chosen operating reflux ratio. The Gilliland X parameter was computed as:

$$X = \frac{\left(\frac{L}{D}\right) - \left(\frac{L}{D}\right)_{\min}}{\left(\frac{L}{D}\right) + 1} \quad \Rightarrow \quad X = 0.00265$$

Using the correlation form for Y:

$$Y = \frac{1 - X^{0.0031}}{1 - 0.99357\, X^{0.0031}} \quad \Rightarrow \quad Y = 0.7428$$

The stage requirement was then obtained from the standard Gilliland relationship:

$$Y = \frac{N - N_{\min}}{N + 1} \quad \Rightarrow \quad N = \frac{Y + N_{\min}}{1 - Y} \quad \Rightarrow \quad N = 18.48 \text{ stages}$$

To convert theoretical stages to actual trays, an overall tray efficiency of $E = 0.80$ was applied:

$$N_{act} = \frac{N}{E} \quad \Rightarrow \quad N_{act} = 23.10 \approx 23 \text{ trays}$$

This tray count is consistent with a practical vacuum fractionation cut and reflects the expected increase in required stages when operating close to minimum reflux.

Column height was estimated using a standard tray spacing of 2 ft and additional allowances for top and bottom equipment/disengagement regions:

$$H = (N_{act} - 1)(2\ ft) + 4\ ft(top) + 8\ ft(bottom) \quad \Rightarrow \quad H = 56\ ft$$

Finally, column diameter was estimated by tray hydraulics using the maximum internal vapor-traffic stage identified in Aspen Hydraulic Plots (Stage 21). Flooding velocity was estimated using the Souders–Brown form with tray capacity constant C, and a design velocity at 80% of flooding was applied:

$$u_f = C\sqrt{\frac{\rho_L - \rho_V}{\rho_V}} \quad \Rightarrow \quad u_f = 3.34\ m/s$$

$$u_{des} = 0.8u_f \quad \Rightarrow \quad u_{des} = 2.67\ m/s$$

Diameter was then computed using the vapor mass flow rate $\dot{m}_V$, vapor density ρ_V, and an active area fraction $\phi = 0.8$to account for tray downcomers and non-active cross-sectional area. A tray capacity constant of $C = 0.11$ m/s was used:

$$D = \sqrt{\frac{4\dot{m}_V}{\pi \rho_V \phi u_{des}}} \quad (\phi = 0.8,\ C = 0.11\ m/s) \quad \Rightarrow \quad D = 4.16\ m = 13.6\ ft$$

The resulting diameter reflects the large cross-sectional area typically required for vacuum towers, where low vapor density increases volumetric flow and hydraulics become controlling. Differences between the hand diameter and Aspen-reported diameter are expected because Aspen tray sizing incorporates additional constraints (e.g., downcomer backup, entrainment, weeping, and tray geometry assumptions) and may be governed by a different internal location depending on the controlling hydraulic limit.

For comparison to Aspen's reported stage count, RADFRAC reports theoretical stages only (excluding condenser and reboiler). Aspen reported 22 theoretical stages for C-401; applying the same overall tray efficiency assumption ($E = 0.80$) gives an estimated physical tray count of $22/0.80 \approx 27$trays, which is consistent in form with the hand-correction procedure used above. The Aspen results and hand calculations are compared in the table below

Table 30: Hand vs Aspen Calculation for Column C-401

C-401	Hand Calculation	Aspen calculation
Column Trays	23 trays	27 trays
Column Diameter	13.6 ft	14.0 ft
Column Height	56 ft	70 ft

As shown in *Table 30*, although Aspen predicts a larger column height than the hand estimate, the simulated sizing results remain consistent with typical industrial vacuum column dimensions and provide a reliable basis for preliminary sizing.

Appendix VIII: Preliminary Utilities

The following, *Table 31*, displays the preliminary utilities needed to operate the process. These specifications were obtained from the process simulated on ASPEN v14.

Table 31: Cooling Water Flowrates

Equipment	Flowrate (gpm)
E-101	68046
E-102	3339
E-201	64182
E-300	37355

E-401	7829
E-400A	23196
E-401A	2209
E-501	17975
E-500A	9589
Total	**233720**

Table 32: Electricity Requirement for Equipment

Equipment	Power (kw)
K-100	5.33
K-200	13204
K-201	11310
P-400	26.72
P-400A	0.023
P-401A	0.049
K-400	14.31
P-500	297

P-501	11
P-500A	0.17
K-500	2180
Total	**27048.6**

Table 33: Steam Requirements for Equipment			
Equipment	Duty (kw)	Flow (lbs/hr)	Type
E-100	0.979	3.54	LP
E-400B	1011.3	3944.6	MP
DRY-100	73751.6	340447.7	HP
E-500	1337.2	6172.6	HP
E-500B	7621.0	35179.8	HP
E-401B	1540.5	7111.1	HP

Table 34: Total Steam Consumption by Type		
Type	Duty (kw)	Flow (lbs/hr)
LP steam	0.979	3.54
MP steam	1011.3	3.944.6

HP steam	84250.3	392859.34

Table 35: Natural Gas Consumption by Equipment		
Equipment	Duty (kw)	Flow (lbs/hr)
R-100	21736.06	107819.5
E-400	1857.551	9214.197
R-101	34166.13	169477.7
Total	**57759.74**	**286511.397**

Appendix IX: Economic Evaluation

This appendix contains the detailed calculations supporting the economic analysis presented in Section 7. While the main report summarizes the key economic indicators and conclusions, the tables included here provide the full intermediate calculations used to determine the net present value (NPV), discounted cash flow rate of return (DCFROR), depreciation, taxable income, taxes, and discounted cash flows for both the MACRS-5 and MACRS-7 depreciation schedules. The appendix also includes the installed cost estimates for all major process equipment. Additionally, the complete datasets used for the sensitivity analysis are provided, showing how variations in capital cost, operating costs, and product revenue influence the final NPV. These detailed calculations are included for completeness and transparency, allowing the economic methodology and results presented in the main report to be fully verified.

years	Pre-tax Cash Flow	Depreciation	Taxable Income	Tax	CF	Fd	PV of CF	NPV
1	-49.0		-49.0	0.0	-49.0	0.9	-42.6	-42.6
2	-147.1		-147.1	0.0	-147.1	0.8	-111.2	-153.9
3	269.9	71.1	198.9	0.0	269.9	0.7	177.5	23.6
4	551.2	121.8	429.4	52.7	498.5	0.6	285.0	308.7
5	551.2	87.0	464.2	113.8	437.4	0.5	217.5	526.1
6	551.2	62.1	489.1	123.0	428.2	0.4	185.1	711.2
7	551.2	44.4	506.8	129.6	421.6	0.4	158.5	869.7
8	551.2	44.4	506.9	134.3	416.9	0.3	136.3	1006.0
9	551.2	44.4	506.8	134.3	416.9	0.3	118.5	1124.5
10	551.2	22.2	529.0	134.3	416.9	0.2	103.1	1227.6
11	551.2	0.0	551.2	140.2	411.0	0.2	88.3	1315.9
12	551.2	0.0	551.2	146.1	405.1	0.2	75.7	1391.7
13	551.2	0.0	551.2	146.1	405.1	0.2	65.8	1457.5
14	551.2	0.0	551.2	146.1	405.1	0.1	57.3	1514.8
15	551.2	0.0	551.2	146.1	405.1	0.1	49.8	1564.6
16	551.2	0.0	551.2	146.1	405.1	0.1	43.3	1607.8
17	551.2	0.0	551.2	146.1	405.1	0.1	37.6	1645.5
18	551.2	0.0	551.2	146.1	405.1	0.1	32.7	1678.2
19	551.2	0.0	551.2	146.1	405.1	0.1	28.5	1706.7
20	551.2	0.0	551.2	146.1	405.1	0.1	24.8	1731.5
21	551.2	0.0	551.2	146.1	405.1	0.1	21.5	1753.0
22	551.2	0.0	551.2	146.1	405.1	0.0	18.7	**1771.7**

Figure 16: NPV analysis for MACRS-7 year

years	Pre-tax Cash Flow	Depreciation	Taxable Income	Tax	CF	Fd	PV of CF	NPV
1	-49.0		-49.0	0.0	-49.0	0.9	-42.6	-42.6
2	-147.1		-147.1	0.0	-147.1	0.8	-111.2	-153.9
3	269.9	99.5	170.5	0.0	269.9	0.7	177.5	23.6
4	551.2	159.1	392.1	45.2	506.0	0.6	289.3	313.0
5	551.2	95.5	455.7	103.9	447.3	0.5	222.4	535.3
6	551.2	57.3	493.9	120.8	430.4	0.4	186.1	721.4
7	551.2	57.3	493.9	130.9	420.3	0.4	158.0	879.5
8	551.2	28.6	522.6	130.9	420.3	0.3	137.4	1016.9
9	551.2	0.0	551.2	138.5	412.7	0.3	117.3	1134.2
10	551.2	0.0	551.2	146.1	405.1	0.2	100.1	1234.3
11	551.2	0.0	551.2	146.1	405.1	0.2	87.1	1321.4
12	551.2	0.0	551.2	146.1	405.1	0.2	75.7	1397.1
13	551.2	0.0	551.2	146.1	405.1	0.2	65.8	1463.0
14	551.2	0.0	551.2	146.1	405.1	0.1	57.3	1520.2
15	551.2	0.0	551.2	146.1	405.1	0.1	49.8	1570.0
16	551.2	0.0	551.2	146.1	405.1	0.1	43.3	1613.3
17	551.2	0.0	551.2	146.1	405.1	0.1	37.6	1651.0
18	551.2	0.0	551.2	146.1	405.1	0.1	32.7	1683.7
19	551.2	0.0	551.2	146.1	405.1	0.1	28.5	1712.2
20	551.2	0.0	551.2	146.1	405.1	0.1	24.8	1736.9
21	551.2	0.0	551.2	146.1	405.1	0.1	21.5	1758.5
22	551.2	0.0	551.2	146.1	405.1	0.0	18.7	**1777.2**

Figure 17: NPV analysis for MACRS-5 year

7-year	net change	ISBL	VCOP	FCOP	Revenue
	-30%	1818	2001	1783	952
	-20%	1802	1925	1780	1225
	-10%	1787	1848	1776	[illegible]
	0%	1772	1772	1772	1772
	10%	1756	1695	1768	2045
	20%	1741	1619	1764	2318
	[illegible]	[illegible]	[illegible]	[illegible]	591

5-year	venue
	958
	1231
	1504
	1777
	2050
	2323
	2597

Figure 18:Sensitivity Analysis Data for both MACRS years

MACRS-7 year (%)	MACRS-5 year (%)
14.29	20
24.49	32
17.49	19.2
12.49	11.52
8.93	11.52
8.92	5.76
8.93	0
4.46	0

Figure 19: Depreciation Values

Name	Installed Cost [USD]	MOC	F_m	New Cost [USD]
K-100	870,600	CS	1	870,600
E-100	59,200	CS	1	59,200
DRY-100	6,333,300	CS	1	6,333,300
R-100	16,000,000	CS	1	16,000,000
CYC-200	4,000,000	CS	1	4,000,000
E-200A	1,883,000	CS	1	1,883,000
E-200B	1,099,900	CS	1	1,099,900
E-200C	246,700	CS	1	246,700
V-200	261,400	CS	1	261,400
E-201	360,400	CS	1	360,400
V-201	270,500	CS	1	270,500
R-200	3,750,000	1.25CR-0.5MO Alloy	1.6	6,000,000
C-200	670,000	CS	1	670,000
E-300A	186,300	CS	1	186,300
E-300B	576,200	CS	1	576,200
K-300A	1,536,632	CS	1	1,536,632
K-300B	1,398,296	CS	1	1,398,296
K-300C	1,247,052	CS	1	1,247,052
R-300	10,000,000	316 S.S	1.3	13,000,000
E-301A	580,200	CS	1	580,200
E-301B	982,500	CS	1	982,500
E-301C	86,200	CS	1	86,200
V-300	229,900	CS	1	229,900
E-302	304,900	CS	1	304,900
V-301	151,800	CS	1	151,800
P-400	128,800	CS	1	128,800
E-400	350,000	CS	1	350,000
R-400	5,000,000	1.25CR-0.5MO Alloy	1.6	8,000,000
K-400	3,000,000	CS	1	3,000,000
E-401	199,200	CS	1	199,200
V-400	185,700	CS	1	185,700
C-400	3,000,000	CS	1	3,000,000
C-401	7,000,000	CS	1	7,000,000
P-500	293,500	CS	1	293,500
E-500	132,300	CS	1	132,300
K-500	7,000,000	CS	1	7,000,000
P-501	272,200	CS	1	272,200
R-500	12,500,000	2.25CR-0.5MO Alloy	1.9	23,750,000
E-501	219,400	CS	1	219,400
V-500	298,600	CS	1	298,600
C-500	1,432,800	CS	1	1,432,800

Figure 20: Installed Costs and MOC of all equipment

CW

Equipment	duty (G/hrJ)	flow (lb/hr)	gpm	Cost ($/hr)
E-100	0.0	40.3	0.1	0.0003
E-301C	4.2	221125.9	441.9	1.5
E-302	35.6	1881933.9	3760.9	12.6
E-501	16.8	888526.8	1775.6	6.0
C-500	9.0	473981.4	947.2	3.2
R-500	12.4	656330.6	1311.6	4.4
E-401	7.4	393189.2	785.7	2.6
C-400	2.1	110946.8	221.7	0.7
C-401	22.5	1188088.8	2374.3	8.0
E-200C	37.2	1965922.2	3928.7	13.2
E-201	3.3	172074.2	343.9	1.2
total	150.5	7952160.1	15891.6	53.3
			$/oper yr	426,302

Natural Gas

Equipment	duty (GJ//hr)	flow (lb/hr)	Cost ($/hr)
E-400	6.7	9229.5	40.2
R-200	121.8	167819.6	730.8
Total	128.5	177049.1	771.0
		$/oper yr	6,167,667

HP steam

Equipment	duty (G/hrJ)	flow (lb/hr)	Cost ($/hr)
C-401	5.51	7068.04	54.19
C-500	27.05	34684.72	265.90
E-500	4.63	5935.21	45.50
C-400	3.65	4680.55	35.88
Total	40.84	52368.51	401.47
		$/oper yr	3,211,723

LP Steam Gen

Equipment	duty (G/hrJ)	flow (lb/hr)	Cost ($/hr)
E-301B	35.1	35337.4	273.3
E-200B	38.0	38250.2	295.9
E-300A	3.1	3086.0	23.9
E-300B	32.0	32224.3	249.3
Total	108.3	108898.0	842.3
		$/oper yr	6,738,630

Electricity

Equipment	duty (kWh)	Cost ($/hr)
K-100	3.9	0.2
K-300A	6589.0	369.6
K-300B	5617.5	315.1
K-300C	4627.5	259.6
P-500	220.0	12.3
P-501	8.2	0.5
K-500	1239.3	69.5
P-400	26.8	1.5
K-400	14.3	0.8
Total	18346.5	1029.2
	$/oper yr	8,233,899

HP Steam Gen

Equipment	duty (G/hrJ)	flow (lb/hr)	Cost ($/hr)
E-301A	21.0	26979.8	206.3
E-200A	561.6	722014.4	5520.3
Total	582.6	748994.1	5726.6
		$/oper yr	45,812,527

Figure 21: Utility Consumption for each equipment

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